

CHAPTER 1: MATHEMATICAL MODELLING

Mathematical modelling is the process of translating complex real-world phenomena into representative mathematical language. It provides a framework for understanding, predicting, and optimizing systems by simplifying reality into manageable, idealized components. This chapter deals with two major aspects: the nature of mathematical models and the systematic process of designing them.

1.1 Mathematical Models

Definition of a Mathematical Model

A **mathematical model** is a simplified abstraction of a real-world situation expressed in mathematical terms (using equations, inequalities, functions, graphs, etc.). The purpose is to capture the essential behavior of a system while omitting extraneous details. In practice, models are used to:

- Forecast trends and outcomes (e.g., population growth, commodity pricing).
- Understand underlying mechanisms (e.g., fluid flow, epidemic spread).
- Optimize decisions in business, science, and engineering.

Example: The **logistic growth model** for populations is given by

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right),$$

where:

- $P(t)$ is the population at time t ,
- r is the intrinsic growth rate, and
- K is the carrying capacity of the environment.

This equation simplifies the myriad factors affecting growth into two main parameters, allowing predictions about population saturation.

Aim of a Mathematical Model

The primary aim of designing a mathematical model is to *include all the main features of the real-world system* so that the model can:

- **Simplify Complexity:** Disentangle or reduce complicated phenomena to key variables and interactions.

- **Make Predictions:** Forecast outcomes under different scenarios.
- **Facilitate Analysis:** Provide insights into how changes in variables affect the result.
- **Drive Decision-Making:** Offer cost-effective and rapid evaluations, support planning, and guide potential modifications to the system or process.

Key Consideration: Because the real world is inherently complex, models are built on a series of assumptions that capture the dominant effects while deliberately neglecting minor influences. This trade-off between realism and simplicity is critical in both the formulation and interpretation of any model.

Advantages of Mathematical Models

1. **Efficiency and Speed:**
 - **Quick to Produce:** Once the governing equations are set, simulations or computations are usually fast, especially with modern computing.
 - **Cost-Effective:** Compared to real-world experimentation, running a model to produce results is much less expensive.
2. **Predictive Power:**
 - **Forecasting Ability:** Models can predict future behavior under given conditions (e.g., economic trends, weather patterns).
 - **Sensitivity Analysis:** They allow exploration of how changes in input variables affect outcomes.
3. **Enhanced Understanding:**
 - **Clarification of Processes:** By reducing a situation to its essential parts, models help reveal underlying mechanisms.
 - **Hypothetical Analysis:** They allow “what-if” scenarios that are impractical or impossible to test directly in the real world.
4. **Simplification of Complex Situations:**
 - Models turn multifaceted problems into a manageable set of equations or logical structures.
 - This simplification can make simulation and optimization feasible, even for very large systems.
5. **Cost Efficiency in Experimentation:**
 - Simulation runs in a virtual model are far less expensive and quicker than large-scale field experiments.

Disadvantages of Mathematical Models

1. **Oversimplification:**
 - **Loss of Detail:** By abstracting the real world, important nuances might be lost.
 - **Assumptions:** Models typically involve assumptions (e.g., linearity, independence) that might not always hold in practice, leading to potential errors or misrepresentations.
2. **Limited Applicability:**
 - The model might work only under specific conditions. When conditions deviate from the model’s assumptions, its predictions may become unreliable.

- **Scalability Issues:** A model created for a small-scale scenario may not extend well to larger or more complex environments.
- 3. **Potential for Error:**
 - **Parameter Sensitivity:** Small errors in parameter estimation can lead to significantly different outcomes.
 - **Simplistic Dynamics:** Real-world systems are often dynamic and subject to random fluctuations that a simple model might not capture.
- 4. **Reliance on Data Quality:**
 - Models are often built and calibrated on experimental or observational data. Poor-quality or insufficient data can diminish the model's accuracy.

Real-World Example: Economic models used to forecast market behavior rely on assumptions such as rational behavior, perfect information, and simplified market dynamics. While these models can offer valuable insights, their inability to capture sudden irrational market shifts or black-swan events (like financial crises) demonstrates the limitations inherent in any simplified mathematical representation.

1.2 Designing a Model

Designing a mathematical model is a systematic process that generally unfolds in several well-defined stages. Effective modelling is iterative, ensuring that the model refines over time to better align with observations.

Seven Stages in Model Design

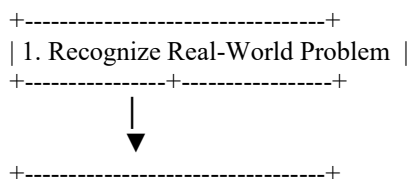
Below is a breakdown of the seven key stages in designing a model, complete with explanations and examples:

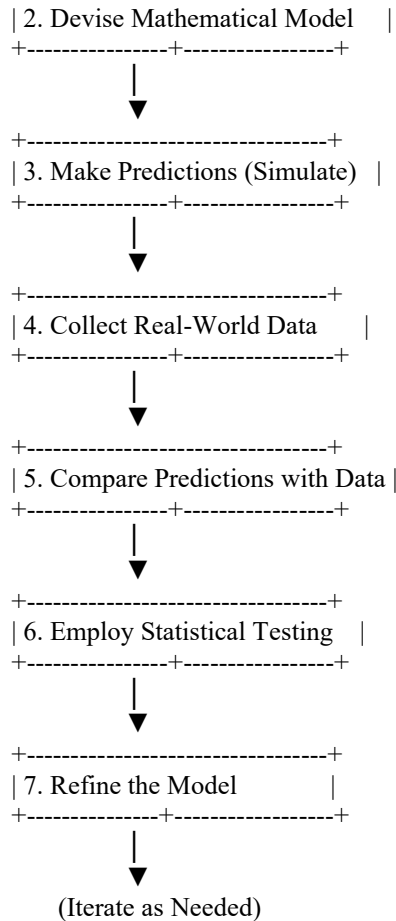
1. **Stage 1: Recognition of a Real-World Problem**
 - **Description:** Identify an issue or phenomenon that requires explanation or prediction.
 - **Example:** Notice that traffic congestion in an urban area is increasing and needs to be analyzed for better management.
2. **Stage 2: Devising a Mathematical Model**
 - **Description:** Translate the real-world problem into mathematical terms. This includes identifying variables, relationships, and making necessary assumptions.
 - **Example:**
 - For traffic flow, one might use models such as the Lighthill-Whitham-Richards (LWR) model.
 - Variables may include vehicle density and flow rate, and assumptions might simplify driver behavior to average reactions.
3. **Stage 3: Using the Model to Make Predictions**
 - **Description:** Once the model structure is established, use it to simulate behavior under various input conditions. This step involves solving the equations or running simulations.
 - **Example:**

- Predict how changes in traffic signal timings might affect congestion.
 - Run simulations that forecast peak traffic volume on future dates.
4. **Stage 4: Collection of Experimental/Real-World Data**
- **Description:** Gather data from observations, experiments, or historical records to provide a benchmark for validating the model.
 - **Example:**
 - Collect traffic data via sensors, cameras, or surveys on vehicle counts and speeds during rush hours.
5. **Stage 5: Comparison Against the Devised Model**
- **Description:** Compare the outcomes predicted by the model with the actual collected data to assess the model's validity.
 - **Example:**
 - Determine whether predicted congestion patterns match what is observed on major roads.
 - Identify discrepancies that suggest missing elements in the model.
6. **Stage 6: Application of Statistical Concepts**
- **Description:** Use statistical tools and tests (such as regression analysis, hypothesis testing, and goodness-of-fit measures) to rigorously assess the model's accuracy and reliability.
 - **Example:**
 - Employ statistical measures like the R-squared value in linear regression to see how well the model predictions explain observed variation in traffic flow.
 - Use hypothesis testing to check whether the deviations are statistically significant compared to measurement errors.
7. **Stage 7: Refinement of the Model**
- **Description:** Based on feedback from data comparison and statistical analyses, modify the model to improve its performance. This can involve adjusting parameters, incorporating additional variables, or revisiting assumptions.
 - **Example:**
 - If the model consistently underestimates traffic density during peak hours, incorporate additional factors such as roadworks or weather conditions, or modify driver reaction parameters.
 - Re-calibrate with new data and repeat the process, gradually improving the model's predictive power.

Visual Representation: The Model Design Process

To help conceptualize the iterative nature of model design, consider the following ASCII flowchart:





Key Takeaways

- **Understanding and Managing Assumptions:** Every mathematical model is built on assumptions. Recognizing which aspects of reality are simplified or idealized is essential for assessing the reliability of a model's predictions.
- **Balance Between Simplicity and Accuracy:** While simple models are advantageous for analysis and computing efficiency, oversimplification can lead to significant errors. The art of modelling lies in striking the right balance.
- **Iterative Improvement:** No model is perfect from the start. The iterative process—collecting data, comparing, and refining—ensures models remain useful as understanding of the system deepens or as conditions change.
- **Broad Applicability:** Mathematical models find applications across many fields—from engineering and economics to epidemiology and environmental science—demonstrating their versatility in problem solving.
- **Real-World Implications:** Understanding the strengths and limitations of mathematical models helps decision-makers interpret predictions critically, ensuring that policies or interventions are based on realistic expectations.

CHAPTER 2: MEASURES OF LOCATION AND SPREAD

Statistical analysis begins by summarizing and understanding data. This chapter explores how we describe the “location” (central tendency and other positions) and the “spread” (variability or dispersion) of data. We start by understanding the types of data, then move on to describe common measures that capture the center and the variability.

2.1 Types of Data

Data come in various forms. Knowing the kind of data you are dealing with is crucial because it dictates which statistical measures and techniques will be appropriate.

Qualitative Variables

- **Definition:** Qualitative variables (also called categorical variables) describe characteristics or attributes that are non-numerical in nature.
- **Examples:**
 - Favourite colour (e.g., blue, red, green)
 - Favourite animal (e.g., dog, cat, elephant)
- **Key Point:** Although these variables may sometimes be coded with numbers (e.g., 1 for blue, 2 for red), the numbers do not have inherent numerical meaning.

Quantitative Variables

- **Definition:** Quantitative variables represent measurable quantities expressed in numbers. They describe “how many” or “how much.”
- **Subdivisions:** They fall into two further categories: **discrete** and **continuous**.

Discrete Data

- **Description:** Discrete data take specific, separate values—often resulting from counting. There are no values “in between.”
- **Examples:**
 - The number of girls in a family
 - The number of bowling pins left standing after a roll
- **Grouped Data:** When data sets are large, discrete data are often summarized in frequency tables:
 - **Classes:** Data are divided into groups or intervals.
 - **Class Boundaries:** The actual dividing points between classes. For example, if a class is listed as 10–19, the class boundaries might be taken as 9.5 to 19.5.

- **Class Midpoint:** The center of an interval, calculated as the average of the lower and upper boundaries.
- **Class Width:** The difference between the upper and lower boundaries of a class interval.

Continuous Data

- **Description:** Continuous data can take on any value within a given range. These are usually measured rather than counted.
- **Examples:**
 - A person's weight (which can be 68.2 kg, 68.25 kg, etc.)
 - The distance between two points (which may be 5.38 km, 5.391 km, and so on)
- **Grouped Presentation:** Often, continuous data are grouped into classes so that measures like the class midpoint can be used in further calculations (for example, estimating the mean).

2.2 Measures of Central Tendency

Measures of central tendency describe the center or “typical” value of a dataset. They provide a single value that attempts to represent a set of data by identifying the central position within that set of data.

Mode (or Modal Class)

- **Definition:** The mode is the value or the class interval that occurs most frequently in a dataset.
- **Application:**
 - **For Raw Data:** Identify the value that appears the most.
 - **For Grouped (Frequency Table) Data:** The modal class is the class with the highest frequency.
- **Example:** Given the data: {3, 5, 8, 5, 7, 5, 9}, the mode is 5 because it occurs three times.
- **Advantages:**
 - Applicable to any type of data (qualitative, quantitative, discrete, or continuous).
 - Useful when the most common category is of interest.

Median

- **Definition:** The median is the middle value when all data are arranged in order.
- **For Raw Data:**
 - If the number of observations, n , is odd: the median is the $\frac{n+1}{2} - th$ value.
 - If n is even: the median is the average of the $\frac{n}{2} - th$ and $(\frac{n}{2} + 1) - th$ values.
- **For Grouped Data:** Interpolation is used to estimate the median because the individual values within each class are not known. The assumption is that data values are evenly distributed within the class.

- **Example:** For the ordered set {2, 4, 6, 8, 10} ($n = 5$), the median is the 3rd value, 6.
- **Advantages:**
 - Less affected by extreme values (outliers) than the mean.
 - Preferred for skewed distributions.

Mean

- **Definition:** The mean (often called the average) is the sum of all data values divided by the number of data values.
- **Formula for Raw Data:**

$$\bar{x} = \frac{\sum x}{n}$$

where $\sum x$ is the sum of all the data points and n is the number of values.

- **For Frequency Tables:** If data are given in a frequency table, the mean is calculated by:

$$\bar{x} = \frac{\sum(x \cdot f)}{\sum f}$$

where x is a data value (or the midpoint for grouped data), f is its frequency, and $\sum f$ is the total number of observations.

- **Combined Data Sets:** For two data sets A (with size n_1 and mean $\bar{x}_1$) and B (with size n_2 and mean $\bar{x}_2$):

$$\bar{x}_{\text{combined}} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

- **Example:** Suppose we have numbers {2, 4, 6, 8}. Then:

$$\bar{x} = \frac{2 + 4 + 6 + 8}{4} = \frac{20}{4} = 5.$$

- **Considerations:** The mean uses every value in the dataset, which makes it sensitive to extreme values. It works best for symmetric distributions without outliers.

Deciding on the Best Measure

- **Mode:** Best for nominal (categorical) data and any time the most frequent observation is of interest.
- **Median:** Ideal for skewed distributions or when there are outliers.
- **Mean:** Most efficient for symmetric distributions when every data point's contribution is relevant.

2.3 Other Measures of Location

Beyond the basic measures of central tendency, there are additional ways to describe the position of data within a distribution.

Quartiles

- **Definition:** Quartiles divide a ranked dataset into four equal parts.
- **Lower Quartile (Q_1):** The value below which 25% of the data fall.
 - **For discrete data:**
 - If n is divisible by 4, Q_1 is the average of the $\frac{n}{4} - th$ and $\left(\frac{n}{4} + 1\right) - th$ values.
 - If not, Q_1 is the $\left(\left\lfloor \frac{n}{4} \right\rfloor + 1\right) - th$ value.
- **Upper Quartile (Q_3):** The value below which 75% of the data fall.
 - **For discrete data:**
 - If n is divisible by 4, Q_3 is the average of the $\frac{3n}{4} - th$ and $\left(\frac{3n}{4} + 1\right) - th$ values.
 - If not, Q_3 is the $\left(\left\lfloor \frac{3n}{4} \right\rfloor + 1\right) - th$ value.
- **Example:** Consider the unsorted data set $\{3, 7, 8, 5, 12, 14, 21, 13, 18\}$. After arranging it in order: $\{3, 5, 7, 8, 12, 13, 14, 18, 21\}$.
 - $n = 9$. The median is the 5th value: 12, so 50% of data lie below 12.
 - For Q_1 : $\left\lfloor \frac{9}{4} \right\rfloor = 2$; thus Q_1 is the 3rd value, 7.
 - For Q_3 : $\left\lfloor \frac{3 \times 9}{4} \right\rfloor = 6$; Q_3 is the 7th value, 14.

Percentiles

- **Definition:** Percentiles split a dataset into 100 equal parts. The p -th percentile (P_p) is the value below which $p\%$ of the data fall.
- **Calculation (For Discrete Data):**
 - For the 10th percentile (P_{10}), locate the $\left(\left\lfloor \frac{10n}{100} \right\rfloor + 1\right) - th$ observation.
 - If $\frac{10n}{100}$ is an integer, the 10th percentile is often taken as the average of that p -th value and the next.
- **For Grouped Continuous Data:** Interpolation is applied because the exact data values within classes are not available. You assume the values are uniformly distributed within the class.
- **Example:** If in a test the 10th percentile score must be determined from a frequency distribution of scores, you would:
 1. Determine the cumulative frequency up to the class that contains the 10th percentile.

2. Use the class boundaries, class width, and cumulative frequency to interpolate the exact score.

2.4 Measures of Spread

Measures of spread (or dispersion) describe how the data values are distributed around the center. They complement measures of central tendency by indicating the variability in the data.

Range

- **Definition:** The range is the difference between the maximum and minimum values in the data set.
- **Formula:**

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

- **Example:** For the data set $\{3, 7, 8, 5, 12\}$, the range is $12 - 3 = 9$.
- **Note:** Although simple to calculate, the range is highly sensitive to extreme values (outliers).

Interquartile Range (IQR)

- **Definition:** The IQR measures the spread of the middle 50% of the data.
- **Formula:**

$$\text{IQR} = Q_3 - Q_1$$

- **Example:** Using the previous quartile example with $Q_1 = 7$ and $Q_3 = 14$,

$$\text{IQR} = 14 - 7 = 7.$$

- **Advantages:** IQR is robust against outliers and gives a better picture of the typical spread for skewed distributions.

Interpercentile Range

- **Definition:** Similar to IQR but based on percentiles. Frequently, the difference between the 10th and 90th percentiles is used.
- **Example:** If the 10th percentile is 15 and the 90th percentile is 85, then the interpercentile range is $85 - 15 = 70$.

Variance and Standard Deviation

While these are expanded in the next section, they are key measures of spread:

- **Variance:** Measures the average of the squared deviations from the mean.
- **Standard Deviation:** The square root of the variance, giving a measure of spread in the original units of the data.

2.5 Variance and Standard Deviation

Understanding the variability around the mean is crucial. Variance and standard deviation provide this information in two different forms.

Variance

- **Definition:** Variance is the average squared deviation of each data point from the mean.
- **For a Sample:**

$$s^2 = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n} \right)^2$$

where $\bar{x}$ is the sample mean and n is the sample size.

- **For Frequency Tables:**

$$s^2 = \frac{\sum fX^2}{n} - \left(\frac{\sum fX}{n} \right)^2$$

Example (Raw Data): Consider data: {2, 4, 6, 8}.

Step 1: Compute $\sum X$ and $\sum X^2$

- $\sum X = 2 + 4 + 6 + 8 = 20$
- $\sum X^2 = 2^2 + 4^2 + 6^2 + 8^2 = 4 + 16 + 36 + 64 = 120$
- Number of data points: $n = 4$

Step 2: Compute Variance

$$s^2 = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n} \right)^2$$

Substituting values:

$$s^2 = \frac{120}{4} - \left(\frac{20}{4} \right)^2$$

$$s^2 = 30 - (5)^2 \quad s^2 = 30 - (5)2$$

$$s^2 = 30 - 25$$

$$s^2 = 5$$

Standard Deviation

- **Definition:** Standard deviation is the square root of the variance. It measures the average amount of dispersion in the same units as the original data.
- **For a Population:**

$$\sigma = \sqrt{\sigma^2}$$

- **For a Sample:**

$$s = \sqrt{s^2}$$

- **Example Continued:** From the previous variance $s^2 = 5$,

$$s = \sqrt{5} \approx 2.236.$$

- **Interpretation:** About 68% of the data (if normally distributed) lie within one standard deviation of the mean.

2.6 Coding

Coding in statistics is a technique used to simplify calculations and reduce the complexity of working with large or unwieldy data values. It involves applying a linear transformation to each data point.

The Linear Transformation

- **General Form:**

$$y = \frac{x - a}{b}$$

where:

- x is the original data value,
- a is a constant shift (often chosen as a convenient reference value),
- b is a scaling factor.

Effect on the Mean

- **Coded Data Mean:** Given the transformation, the mean of the new coded data is:

$$\bar{y} = \frac{\bar{x} - a}{b}.$$

- **Recovering the Original Mean:** Rearranging the formula provides:

$$\bar{x} = a + b\bar{y}.$$

Effect on the Standard Deviation

- **Coded Data Standard Deviation:** Since adding or subtracting a constant does not affect spread, only the scaling factor b impacts standard deviation:

$$\sigma_y = \frac{\sigma_x}{|b|}.$$

- **Reversing the Transformation:**

$$\sigma_x = |b|\sigma_y.$$

- **Example:** Suppose you have a dataset with large values (e.g., incomes in dollars) and choose to code them with $a = 1000$ and $b = 100$. Then:
 - A data point $x = 3500$ is recoded as:

$$y = \frac{3500 - 1000}{100} = 25.$$

- If the coded mean $\bar{y} = 20$ and coded standard deviation $\sigma_y = 3$, then:
 - The original mean is:

$$\bar{x} = 1000 + 100(20) = 3000,$$

- The original standard deviation is:

$$\sigma_x = 100 \times 3 = 300.$$

Summary and Key Takeaways

1. **Types of Data:**
 - **Qualitative:** Non-numeric; used to classify observations (e.g., colour, animal).
 - **Quantitative:** Numeric; subdivided into discrete (countable values) and continuous (measurable values).
2. **Measures of Central Tendency:**
 - **Mode:** Most frequent value (or class); useful for any data type.
 - **Median:** Middle value in an ordered set; robust against outliers.
 - **Mean:** Average value; incorporates every data point but is sensitive to extremes.

3. **Other Measures of Location:**
 - **Quartiles:** Divide data into four parts to capture the distribution's spread.
 - **Percentiles:** Divide data into 100 parts; useful for detailed positional analysis.
4. **Measures of Spread:**
 - **Range:** Simple and quick, but highly affected by outliers.
 - **Interquartile Range (IQR):** Focuses on the central 50%; resistant to skew.
 - **Interpercentile Range:** Often used to understand the spread of the majority of the data.
 - **Variance and Standard Deviation:** Quantitative measures showing how far data values lie around the mean. Variance uses squared deviations; standard deviation provides dispersion in original units.
5. **Coding:**
 - A linear transformation (shifting and scaling) that simplifies calculations.
 - Mean and standard deviation transform predictably under coding, making it easier to handle large data values.

CHAPTER 3: REPRESENTATIONS OF DATA

Data representation is essential for summarizing, understanding, and comparing complex datasets. This chapter covers several visual and numerical methods that help convert raw data into insight. We explore histograms, outliers, box plots, stem–leaf diagrams, skewness, and data comparison techniques.

3.1 Histograms

Histograms are one of the most common ways to graphically represent grouped continuous data, helping you see the underlying distribution.

Key Concepts & Construction

- **Grouped Continuous Data:** Continuous data (e.g., heights, weights, times) are usually measured on a continuous scale. To work with such data, we typically group them into classes (or intervals).
- **Bar Representation:** A histogram consists of adjacent bars. Each bar represents a class:
 - **Width of the Bar:** Equals the *class width* (e.g., if your class spans 10–20, the width is 10).
 - **Height of the Bar:** Chosen so that the **area** (not solely the height) is proportional to the frequency of the class.
- **Equal Class Widths:** When every class interval is of equal width, the height of each bar is directly proportional to the frequency:

Height $\propto$ Frequency.

Example: If three classes (all of width 10) have frequencies 15, 20, and 25, then the corresponding bar heights can simply be 15, 20, and 25.

- **Unequal Class Widths and Frequency Density:** With unequal intervals, a bar's direct height cannot represent frequency because wider classes would "appear" larger and mislead the viewer. Instead, we use **frequency density** defined as:

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$$

When plotted, the vertical axis is scaled in frequency density. The **area** of each bar is then:

$$\text{Area} = \text{Frequency Density} \times \text{Class Width} \Rightarrow \frac{\text{Frequency}}{\text{Class Width}} \times \text{Class Width} = \text{Frequency}.$$

Illustrative Example: Suppose you have two classes:

- Class 1: 0–10 with frequency = 20. Frequency density = $\frac{20}{10} = 2$.
- Class 2: 10–25 with frequency = 30. Width = 15; frequency density = $\frac{30}{15} = 2$.

Even though the class widths differ, each bar's area accurately reflects its frequency.

- **Frequency Polygon:** A frequency polygon provides an alternative, line-based representation of the distribution.
 - **How to Construct It:**
 1. Compute the midpoint of each class.
 2. Plot a point at each midpoint with a vertical coordinate equal to its frequency (or frequency density, as needed).
 3. Connect the points with straight lines.

Visual Idea: If your midpoints are at $x = 5, 15, 25$ with frequencies 20, 30, 25, drawing lines between these gives an approximate outline of your histogram's shape.

3.2 Outliers

Outliers are data points that deviate markedly from the overall pattern. They can indicate special cases, errors, or rare events.

What Are Outliers?

- **Definition:** An outlier is an extreme value that lies outside the "typical" pattern of the data.

- **Quartile-Based Identification:** A common method is to use the interquartile range (IQR). For a dataset with lower quartile Q_1 and upper quartile Q_3 :
 - **Lower Outlier Criterion:** Any value less than:

$$Q_1 - k(Q_3 - Q_1)$$

- **Upper Outlier Criterion:** Any value greater than:

$$Q_3 + k(Q_3 - Q_1)$$

Here, k is a constant:

- **Common Choices:**
 - $k = 1$ or $k = 1.5$ (the latter defines the "inner fences").
 - Values beyond $Q_3 + 1.5 \times (Q_3 - Q_1)$ or below $Q_1 - 1.5 \times (Q_3 - Q_1)$ are commonly marked as outliers.

Example: Suppose $Q_1 = 10$ and $Q_3 = 20$. Then

- $IQR = 20 - 10 = 10$.
- With $k = 1.5$:
 - Lower boundary $= 10 - 1.5 \times 10 = -5$.
 - Upper boundary $= 20 + 1.5 \times 10 = 35$.

Any data point below -5 or above 35 is flagged as an outlier.

- **Graphical Identification:** Outliers are easily spotted in box plots (see Section 3.3) and scatter plots.
- **Cleaning Data:** The process of identifying and possibly removing or correcting these anomalies is known as **cleaning the data**. It ensures that subsequent analyses are not unduly influenced by these extreme values.

3.3 Box Plots

Box plots (or box-and-whisker diagrams) offer a compact picture of a data set's distribution, highlighting the central tendency, spread, and any outliers.

Structure of a Box Plot:

- **The Box:**
 - **Lower Boundary:** Q_1 (25th percentile)
 - **Upper Boundary:** Q_3 (75th percentile)
 - **Inside the Box:** A line representing the median (50th percentile, Q_2).
- **Whiskers:**
 - Extend from the box to the smallest and largest values that are *not* outliers.

- Limits of whiskers are typically defined as:

$$\text{Lower Whisker} \geq Q_1 - 1.5 \times IQR$$

$$\text{Upper Whisker} \leq Q_3 + 1.5 \times IQR$$

where $IQR = Q_3 - Q_1$.

- **Outliers:** Any data values outside the whiskers (beyond $Q_1 - 1.5 \times IQR$ or $Q_3 + 1.5 \times IQR$) are marked with a symbol such as a dot or cross.

Example in Practice:

Consider a dataset where:

- $Q_1 = 15$,
- *Median* = 20,
- $Q_3 = 25$. Then $IQR = 25 - 15 = 10$.
- Lower fence = $15 - 1.5(10) = 0$.
- Upper fence = $25 + 1.5(10) = 40$.

Any data point below 0 or above 40 is treated as an outlier. A drawn box plot would show:

- A box from 15 to 25,
- A median line at 20,
- Whiskers stretching to the minimum and maximum values within $[0, 40]$,
- Outliers plotted individually outside these whiskers.

Why Box Plots?

- **Comparison:** They allow you to quickly compare the central tendency, range, variability, and presence of outliers among multiple datasets.
- **Visualization:** A box plot succinctly displays essential summary statistics.

3.4 Stem and Leaf Diagrams

Stem and leaf diagrams provide a way to display numerical data while preserving the original values. They offer both a quantitative summary and a visual representation of data distributions.

How to Construct a Stem and Leaf Diagram

- **Separation of Data:** Each data value is divided into the **stem** and **leaf**:
 - **Stem:** Usually comprises the leading digit(s) (e.g., tens or hundreds).
 - **Leaf:** Typically the final digit.
- **Example:** Consider the dataset:

23, 27, 31, 34, 36, 41, 45, 47, 52

Create the diagram by splitting each number:

```
Stem | Leaf
2   | 3 7
3   | 1 4 6
4   | 1 5 7
5   | 2
```

A **key** like “2 | 3 = 23” must be provided to clarify the representation.

- **Back-to-Back Stem and Leaf Diagrams:** These are constructed to compare two datasets on the same scale by listing the stems in a common column and placing the leaves for one dataset on the left and the leaves for the other on the right.

Sketch Example:

Dataset A		Stem		Dataset B
7 8		4		1 3 5
2 4 6		3		7 8

This side-by-side layout facilitates direct comparison of distributions.

3.5 Skewness

Skewness measures the asymmetry in a distribution of data. Understanding skewness helps assess whether your data are balanced or lopsided.

Types of Skewness

- **Symmetrical Distribution:** When data are evenly distributed:

$$\text{Mean} \approx \text{Median} \approx \text{Mode}.$$

In a box plot, the median is centered within the box, and the whiskers are roughly equal in length.

- **Positive Skew (Right-Skewed):**
 - **Characteristics:** The tail extends to the right; the mean is greater than the median.

$$\text{Mean} > \text{Median} > \text{Mode}.$$

- **In Box Plots:** The median is closer to Q_1 (the lower quartile) and the right whisker is longer.
- **Negative Skew (Left-Skewed):**

- **Characteristics:** The tail extends to the left; the mean is less than the median.

$$\text{Mean} < \text{Median} < \text{Mode}.$$

- **In Box Plots:** The median is closer to Q_3 (the upper quartile) and the left whisker is extended.

Calculating Skewness

A common formula used is:

$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

- **Interpretation:**
 - A **positive value** indicates a right-skewed distribution.
 - A **negative value** indicates a left-skewed distribution.
 - A value **close to zero** suggests the data are approximately symmetrical.

Example: Suppose:

- Mean = 50, Median = 45, and Standard Deviation = 10.

$$\text{Skewness} = \frac{3(50 - 45)}{10} = \frac{15}{10} = 1.5.$$

A skewness of 1.5 indicates a positively skewed distribution.

3.6 Comparing Data

When comparing data sets, you must examine both numerical summaries and graphical displays to understand differences in central tendency, spread, and shape.

Key Elements in Data Comparison

1. **Measures of Location:**
 - **Mean:** Provides the average value.
 - **Median:** Offers a robust center less affected by outliers.
 - **Mode:** Indicates the most frequent observation.
2. **Measures of Spread:**
 - **Standard Deviation/Variance:** Quantify the dispersion of data values.
 - **Interquartile Range (IQR):** The spread of the middle 50% of the data.
 - **Range:** The difference between the extreme values.
3. **Graphical Representations:**
 - **Histograms:** Compare overall frequency distributions.

- **Box Plots:** Quickly reveal differences in quartiles, skew, and outliers.
- **Stem and Leaf Diagrams:** Retain exact values and offer an honest look at the underlying numbers.

Example of Data Comparison:

Imagine you wish to compare exam scores for two classes:

- **Class A:**
 - Mean = 70, Median = 72, IQR = 10.
 - Histogram shows a nearly symmetric, bell-shaped distribution.
 - Box plot reveals a centered median and few or no outliers.
- **Class B:**
 - Mean = 65, Median = 60, IQR = 15.
 - Histogram indicates a slight left skew.
 - Box plot reveals a shifted median with a longer left whisker and a few low-value outliers.

The joint analysis shows that Class A not only has a higher central tendency but also exhibits less variability compared to Class B.

Summary

- **Histograms:**
 - Visualize grouped continuous data as adjacent bars where the area of each bar (height $\times$ width) represents the frequency.
 - For unequal class widths, use frequency density ($\frac{\text{frequency}}{\text{class width}}$) to ensure correct area representation.
 - A frequency polygon can be added by connecting class midpoints.
- **Outliers:**
 - Extreme data values identified using quartile fences (e.g., $Q_1 - k(Q_3 - Q_1)$ and $Q_3 + k(Q_3 - Q_1)$).
 - Values beyond the inner fences (usually $k = 1.5$).
 - Removing such anomalies is part of the data cleaning process.
- **Box Plots:**
 - Summarize distribution via the median, quartiles, and potential outliers.
 - Whiskers extend to non-outlier extreme values.
 - Compare multiple datasets easily.
- **Stem and Leaf Diagrams:**
 - Display original data values via stems (leading digits) and leaves (trailing digits).
 - Provide a clear view of data distribution and are useful for small to medium-sized datasets.
 - Back-to-back displays facilitate comparisons between datasets.
- **Skewness:**

- Measures asymmetry: positive skew (tail on the right), negative skew (tail on the left).
 - Quantified by the formula $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$.
 - Aids in understanding distribution shape and potential bias in the mean relative to the median.
- Comparing Data:**
 - Use a combination of measures of location (mean, median, mode) and spread (standard deviation, IQR, range) to assess datasets.
 - Graphical methods like histograms and box plots are invaluable in these comparisons.

CHAPTER 4: PROBABILITY

Probability is the mathematical study of randomness and uncertainty. It provides tools to model chance experiments, quantify the likelihood of events, and make forecasts based on incomplete information. In this chapter, we explore the basic vocabulary, graphical tools, key properties, and rules that govern probability calculations.

4.1 Understanding the Vocabulary Used in Probability

Before we begin calculating probabilities, it is essential to master the terminology:

- Experiment:** A *repeatable* process that produces a set of outcomes. *Example:* Tossing a coin, rolling a die, or drawing a card from a deck.
- Outcome:** A single possible result of an experiment. *Example:* When rolling a die, one outcome could be “4.”
- Event:** A collection (set) of one or more outcomes. *Example:*
 - Getting an even number when rolling a die: The event is $\{2, 4, 6\}$.
 - Drawing a heart from a deck of cards: The event consists of 13 outcomes (one for each heart).
- Sample Space (ξ or S):** The set of all possible outcomes of an experiment. *Example:*
 - For rolling one die: $S = \{1, 2, 3, 4, 5, 6\}$.
 - For tossing a coin twice: $S = \{HH, HT, TH, TT\}$.
- Probability:** A measure of the likelihood of an event occurring. It is always a number between 0 and 1 (inclusive), where 0 means the event cannot occur and 1 means it is certain. *Key Fact:*

$$\sum_{\text{all outcomes in } S} P(\text{outcome}) = 1.$$

When outcomes are *equally likely*, each outcome $x \in S$ has:

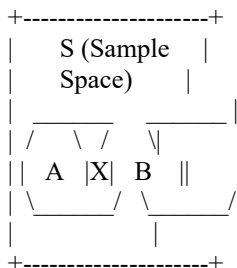
$$P(x) = \frac{1}{|S|}.$$

4.2 Venn Diagrams

Venn diagrams provide a visual way to organize and display the relationships among events within the sample space.

- **Basic Structure:**
 - The overall rectangle represents the sample space ξ (or S).
 - Circles (or ellipses) within the rectangle represent individual events.
 - Overlapping areas indicate the intersection (common outcomes) of events.
- **Key Components:**
 - **Intersection ($A \cap B$):** The overlapping region between circles for events A and B shows outcomes common to both events.
 - **Union ($A \cup B$):** The total area covered by both circles (including their overlap) represents outcomes that belong to A , B , or both.
 - **Complement (A' or A^c):** The region inside the rectangle but outside the circle for event A represents all outcomes where A does not occur.
- **Usage:** Venn diagrams are especially useful to visualize problems involving two or more events (such as calculating $P(A \cup B)$ or verifying if events are mutually exclusive).

Example Diagram:



Here, the area marked “X” represents $A \cap B$, the overlap of events A and B .

4.3 Mutually Exclusive and Independent Events

Understanding the relationship between events is vital in probability.

Mutually Exclusive Events

- **Definition:** Events A and B are *mutually exclusive* (or disjoint) if they cannot occur simultaneously.

$$A \cap B = \emptyset.$$

- **Probabilistic Consequences:**

- Since they share no outcomes, the probability of both events occurring at the same time is zero:

$$P(A \cap B) = 0.$$

- The probability of either AA or BB occurring is given by:

$$P(A \text{ or } B) = P(A) + P(B).$$

- *Example:* When rolling a die, the event $A = \{\text{rolling a 2}\}$ and $B = \{\text{rolling a 5}\}$ are mutually exclusive.

Independent Events

- **Definition:** Two events A and B are *independent* if the occurrence of one does not affect the probability of the occurrence of the other.
- **Mathematical Condition:**

$$P(A \cap B) = P(A) \times P(B).$$

- *Alternate Check:* If $P(B) > 0$, then A and B are independent if:

$$P(A|B) = P(A).$$

- *Example:* Tossing two coins: The outcome of the first toss (e.g., heads) does not affect the outcome of the second toss. So, the probability of getting two heads is:

$$P(\text{H on 1st AND H on 2nd}) = P(\text{H on 1st}) \times P(\text{H on 2nd}) = 0.5 \times 0.5 = 0.25.$$

Note: Mutually exclusive events cannot be independent (except in trivial cases) because if one occurs, the other cannot occur.

4.4 Set Notation

Set notation is the language used to describe events and their relationships.

- **Intersection ($A \cap B$):** Represents outcomes common to both events AA and BB.
- **Union ($A \cup B$):** Represents outcomes that belong to AA, BB, or both.
- **Complement (A' or A^c):** Represents outcomes where event AA does not occur.

$$P(A') = 1 - P(A).$$

- **Empty Set ($\emptyset$):** Denotes an event with no outcomes;

$$P(\emptyset) = 0.$$

- **Mutually Exclusive Events:** If A and B are mutually exclusive, then:

$$A \cap B = \emptyset.$$

Example: For a deck of cards, let A be the event “drawing a queen” and B be “drawing a king.” Since a single card cannot be both,

$$A \cap B = \emptyset \quad \text{and} \quad P(A \cup B) = P(A) + P(B).$$

4.5 Addition Rule for Probabilities

The addition rule is used to find the probability that at least one of two events occurs.

- **General Rule:** For any two events AA and BB,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- **Special Case – Mutually Exclusive Events:** When AA and BB cannot occur together,

$$P(A \cup B) = P(A) + P(B),$$

since $P(A \cap B) = 0$.

Example: If $P(A) = 0.3$ and $P(B) = 0.4$ but they have an overlap $P(A \cap B) = 0.1$ then:

$$P(A \cup B) = 0.3 + 0.4 - 0.1 = 0.6.$$

4.6 Conditional Probability

Conditional probability measures the likelihood of an event occurring given that another event has already occurred.

- **Definition:** The probability of event AA given that event BB has occurred is expressed as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{provided } P(B) > 0).$$

Similarly,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad (\text{provided } P(A) > 0).$$

- **Multiplication Rule:** Rearranging the conditional probability formula yields:

$$P(A \cap B) = P(A|B) \times P(B) = P(B|A) \times P(A).$$

Example: Suppose in a class 40% of the students play basketball and 25% both play basketball and soccer. Then the probability that a student plays soccer given that they play basketball is:

$$P(\text{Soccer}|\text{Basketball}) = \frac{P(\text{Soccer and Basketball})}{P(\text{Basketball})} = \frac{0.25}{0.40} = 0.625.$$

4.7 Probability Formulae

Let's summarize the most important probability rules:

1. **Addition Rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

2. **Multiplication Rule (Conditional Probability):**

$$P(A \cap B) = P(A|B) \times P(B) = P(B|A) \times P(A)$$

3. **Independent Events:** If AA and BB are independent, then:

$$P(A \cap B) = P(A) \times P(B)$$

4. **Conditional Probability:**

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

These rules form the backbone of calculating probabilities in more complex scenarios.

4.8 Tree Diagrams

A tree diagram is a graphical representation of the outcomes of sequential events and their probabilities.

Components & Construction

- **Branches:** Each branch represents a possible outcome at a given stage.

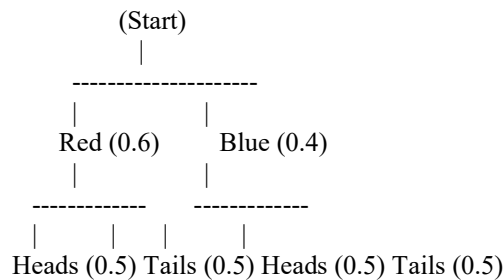
- **Sequential Representation:** Start with an initial branch (or “trunk”) and then draw branches emanating from it for subsequent events. Multiply along the branches to determine the probability of any complete path.
- **Normalization:** At any stage, the sum of the probabilities on the branches emanating from the same node must equal 1.

Example:

Consider a two-step experiment where you first choose a colored ball from a bag (say, Red or Blue) and then flip a coin.

- Let $P(\text{Red}) = 0.6$ and $P(\text{Blue}) = 0.4$.
- For a coin toss, $P(\text{Heads}) = 0.5$ and $P(\text{Tails}) = 0.5$.

The tree diagram would be:



- **Calculations:**
 - Probability of “Red and Heads”: $0.6 \times 0.5 = 0.3$.
 - Probability of “Blue and Tails”: $0.4 \times 0.5 = 0.2$.

Tree diagrams are especially powerful when dealing with conditional events and multiple stages.

Summary

- **Vocabulary:** Understand basic terms—experiment, outcome, event, sample space, probability—and the principle that the sum of all outcome probabilities equals 1.
- **Venn Diagrams:** Graphically display events, their intersections, unions, and complements.
- **Mutually Exclusive vs. Independent Events:**
 - Mutually exclusive events cannot occur together, so $P(A \cap B) = 0$.
 - Independent events do not affect each other’s probabilities, so $P(A \cap B) = P(A) \times P(B)$.
- **Set Notation:** Use symbols like $A \cap B$, $A \cup B$, and A' to compactly describe events and their relationships.
- **Addition and Multiplication Rules:** Fundamental probability formulae for combining events and calculating joint probabilities.

- **Conditional Probability:** Quantifies the probability of an event given the occurrence of another, leading naturally to the multiplication rule.
- **Tree Diagrams:** Visual tools to map sequential events, where probabilities multiply along branches.

CHAPTER 5: CORRELATION AND REGRESSION

In many problems, we are interested in studying the relationship between two quantitative variables. Correlation and regression are two key concepts used to describe and model such relationships. This chapter covers scatter diagrams, linear regression models, the least squares method for fitting a line, the product moment correlation coefficient, and how to interpret these results.

5.1 Scatter Diagrams

A **scatter diagram** (or scatter plot) is a graphical tool that displays pairs of numerical data (bivariate data). Each point on the plot corresponds to one pair of values—one for the independent (explanatory) variable and one for the dependent (response) variable.

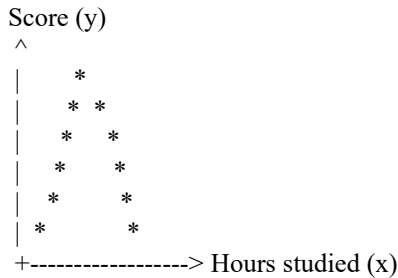
Key Points:

- **Bivariate Data:** Each plotted point (x, y) represents a measurement where:
 - x : independent (explanatory) variable (horizontal axis)
 - y : dependent (response) variable (vertical axis)
- **Purpose:**
 - To visually assess whether there is any relationship between the two variables.
 - To get a first glimpse of the direction (positive/negative) and strength (tight or dispersed clustering) of any linear relationship.
- **Visual Patterns and Correlation:**
 - **Strong Positive Correlation:** Points cluster closely around a line with a positive slope. *Example:* Consider points that lie roughly along the line $y = 2x + 1$.
 - **Weak Positive Correlation:** Points have a general upward trend but display considerable scatter.
 - **No (or Zero) Linear Correlation:** Points are randomly scattered with no apparent upward or downward trend.
 - **Weak Negative Correlation:** Points show a general downward trend but are spread out.
 - **Strong Negative Correlation:** Points cluster closely around a line with a negative slope (e.g., $y = -3x + 10$).

Example:

Imagine recording the number of hours studied (x) and the resulting test scores (y) for a group of students. A scatter diagram with hours on the x-axis and scores on the y-axis might show that as the number of study hours increases, the test score generally increases—perhaps forming a pattern that suggests a positive linear relationship.

Visual Illustration (Simplified ASCII Diagram):



In the above sketch, points seem to follow an upward trend, suggesting a positive correlation between study hours and scores.

5.2 Linear Regression

Linear regression is a statistical method used to model the relationship between two variables by fitting a linear equation to observed data. The goal is to create a model that predicts the dependent variable y from the independent variable x .

The Regression Model:

- **General Equation:**

$$y = a + bx \quad \text{or} \quad y = mx + c$$

where:

- y is the **dependent (response) variable**
- x is the **independent (explanatory) variable**
- a (or c) is the **y-intercept**—the value of y when $x = 0$
- b (or m) is the **slope (gradient)**—indicating the change in y for a unit change in x
- **Least Squares Regression:** The most common method for finding the best-fitting line is the **least squares method**, which minimizes the sum of the squares of the vertical differences (residuals) between the observed values and the values predicted by the line.

Why Use Linear Regression?

- It allows for the prediction of new y -values given new x -values.
- It quantifies the strength and nature of the linear relationship.

- The slope gives the rate of change: If $b = 5$, then for each additional unit of x , y increases by 5 units.
- The intercept a grounds the line in the coordinate plane, representing the starting value of y .

5.3 Calculating Least Squares Linear Regression

This section explains in detail how to compute the slope and intercept for the least squares regression line using summary statistics.

Summary Statistics Required:

- Σx : Sum of all x -values
- Σy : Sum of all y -values
- Σx^2 : Sum of the squares of the x -values
- Σy^2 : Sum of the squares of the y -values
- Σxy : Sum of the product of x and y for each data pair
- n : Total number of data pairs

Key Quantities:

- S_{xx} (Sum of squares for x):

$$S_{xx} = \sum (x - \bar{x})^2 = \Sigma x^2 - \frac{(\Sigma x)^2}{n}$$

- S_{xy} (Sum of products):

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}$$

Formulas for Regression Coefficients:

- Slope (b):

$$b = \frac{S_{xy}}{S_{xx}}$$

- Intercept (a):

$$a = \bar{y} - b\bar{x}$$

Worked Example:

Suppose we have data for 5 data points with the following summary statistics:

- $\Sigma x = 25, \Sigma y = 60$
- $\Sigma x^2 = 135$
- $\Sigma xy = 320$
- $n = 5$

1. **Calculate the Means:**

$$\bar{x} = \frac{\Sigma x}{n} = \frac{25}{5} = 5, \quad \bar{y} = \frac{\Sigma y}{n} = \frac{60}{5} = 12.$$

2. **Compute S_{xx} :**

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 135 - \frac{25^2}{5} = 135 - \frac{625}{5} = 135 - 125 = 10.$$

3. **Compute S_{xy} :**

$$S_{xy} = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 320 - \frac{(25)(60)}{5} = 320 - \frac{1500}{5} = 320 - 300 = 20.$$

4. **Determine the Slope b :**

$$b = \frac{S_{xy}}{S_{xx}} = \frac{20}{10} = 2.$$

5. **Determine the Intercept a :**

$$a = \bar{y} - b\bar{x} = 12 - (2 \times 5) = 12 - 10 = 2.$$

Thus, the least squares regression line is:

$$y = 2 + 2x.$$

5.4 The Product Moment Correlation Coefficient

The **product moment correlation coefficient (PMCC)**, denoted by r , is a measure that quantifies the linear relationship between two variables.

Definition and Formula:

- **Formula Using Deviations:**

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

Alternate Formulation:

$$r = \frac{\Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}}{\sqrt{\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right)\left(\Sigma y^2 - \frac{(\Sigma y)^2}{n}\right)}}$$

Interpretation:

- r = ranges between -1 and +1:
 - $r = +1$: Perfect positive linear correlation.
 - $r = -1$: Perfect negative linear correlation.
 - $r = 0$: No linear correlation.
- **Strong Correlation:** Values close to ± 1 .
- **Weak Correlation:** Values near 0.
- **Symmetry:** The correlation between x and y is the same as the correlation between y and x .

Important Note:

- **Correlation Does Not Imply Causation:** Even if two variables have a strong correlation, this does not necessarily mean that one causes the other.

5.5 Interpretation of Correlation and Regression

Once the regression model is developed and the correlation coefficient r is computed, you must be able to interpret what these numbers mean in context.

Interpreting the Product Moment Correlation Coefficient (rr):

- **Direction:**
 - A positive r indicates that as x increases, y tends to increase.
 - A negative r indicates that as x increases, y tends to decrease.
- **Strength:**
 - r close to +1 or -1 implies a strong linear relationship.
 - r near 0 implies little to no linear relationship.

Interpreting the Regression Line $y = a + bx$:

- **Slope (b):** Indicates the change in the response variable y for a one-unit change in the explanatory variable x . *Example:* A slope of 2 means that for each additional unit increase in x , y increases by 2 units.
- **Intercept (a):** Represents the predicted value of y when $x = 0$. *Caveat:* If $x = 0$ is outside the range of observed data, the intercept may not have a meaningful real-world interpretation.

- **Prediction:**
 - **Interpolation:** Predicting y for x values within the range of the observed data is generally reliable.
 - **Extrapolation:** Predicting beyond the observed range of x -values can be unreliable because the relationship may not hold outside the data's boundaries.

Limitations and Considerations:

- **Linearity Assumption:** Linear regression assumes a linear relationship. Non-linear relationships require different models.
- **Influence of Outliers:** Outliers can affect both the slope of the regression line and the value of r .
- **Dependent vs. Independent Variables:** The analysis is based on the assumption that one variable (the dependent or response variable) is influenced by or can be predicted using the other (the independent or explanatory variable).

Practical Scenario:

Imagine measuring the amount of fertilizer applied (x) and the yield of crops (y). A scatter diagram might show a strong positive correlation. Using linear regression, you calculate a regression line:

$$y = 2 + 3x.$$

- This means that for every unit of fertilizer applied, the crop yield increases by 3 units.
- If $r = 0.95$, the relationship is very strong.
- It is safe to use the regression line to predict crop yield for fertilizer amounts within the observed range (interpolation), but caution is warranted if predicting for unusually high or low fertilizer levels (extrapolation).

Summary of Chapter 5

- **Scatter Diagrams:** Provide a visual layout for bivariate data to assess the potential linear relationship, including strength and direction (positive, negative, or zero correlation).
- **Linear Regression:** Constructs a linear model ($y = a + bx$) that predicts the dependent variable from the independent variable using the least squares method.
- **Least Squares Linear Regression Calculations:** Involves computing summary statistics (Σx , Σy , Σx^2 , Σxy), then finding S_{xx} and S_{xy} , and finally determining the slope b and intercept a .
- **Product Moment Correlation Coefficient:** Measures the strength and direction of the linear relationship with r ranging from -1 to $+1$. Recall: Correlation does not equate with causation.
- **Interpretation:** Evaluate the regression line's slope and intercept, and consider the distinction between interpolation and extrapolation. Understand that strong correlation

indicates a likely linear relationship but always check for limitations, such as non-linearity and the influence of outliers.

CHAPTER 6: DISCRETE RANDOM VARIABLES

Discrete random variables arise in experiments where outcomes can only take on a countable number of values. In this chapter, we explore their definitions, probability distributions, cumulative functions, expected values, variance, and how linear functions of these variables behave. We also study the discrete uniform distribution and see how these ideas are applied in problem solving.

6.1 Discrete Random Variables

Definition of a Random Variable

- **Random Variable:** A **random variable** is a function that associates a real number with each outcome in a sample space of an experiment. It is denoted by a capital letter (commonly X). *Example:* In an experiment where you toss a coin twice, you might define X as the number of heads obtained.
- **Key Point:** A random variable translates qualitative outcomes into numerical values that can be analyzed statistically.

Range of Values a Random Variable Can Take

- **Range:** The set of all possible numerical values that a random variable can assume. For discrete random variables, these are countable values. *Example:* A die roll defines X with range $\{1, 2, 3, 4, 5, 6\}$.

Discrete Values

- **Discrete Values:** These are individual, separated numbers (often integers) with no intermediate values between them. *Example:* The number of defective items in a production batch is discrete $(0, 1, 2, \dots)$.

Sample Space

- **Sample Space (S):** The set of all possible outcomes of an experiment. Every outcome where the random variable is defined is a member of this space. *Example:* For the roll of a six-sided die,

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Probability Distribution

- **Probability Distribution:** A probability distribution for a discrete random variable assigns a probability $P(X = x)$ to each value x in the range. The fundamental requirements are:
 1. $0 \leq P(X = x) \leq 1$ for each x .
 2. The sum of the probabilities over all possible x is 1:

$$\sum_x P(X = x) = 1$$

Representations of a Probability Distribution

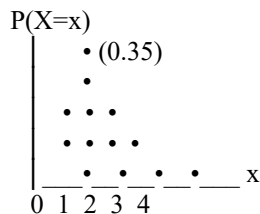
- **Using a Table:** A table is one of the simplest ways to display the probability distribution. Each row lists a value x and its corresponding probability $P(X = x)$.

Example Table:

In this table, all probabilities add up to 1.

- **Using a Diagram:** A probability distribution diagram (sometimes called a probability mass function plot) shows each value x on the horizontal axis, with a vertical “bar” or “dot” whose height represents $P(X = x)$.

Simple Diagram Sketch:



- **Discrete Uniform Distribution:** A special case occurs when every value in the range is equally likely. In a **discrete uniform distribution**, if there are n outcomes, then:

$$P(X = x) = \frac{1}{n} \quad \text{for each } x.$$

Example: A fair die gives

$$P(X = x) = \frac{1}{6}, \quad x \in \{1, 2, 3, 4, 5, 6\}.$$

6.2 Finding the Cumulative Distribution Function for a Discrete Random Variable

The cumulative distribution function (CDF) gives the probability that the random variable takes on a value less than or equal to a specified number.

Definition of Cumulative Distribution Function ($F(x)$)

- **CDF, $F(x)$:**

$$F(x) = P(X \leq x)$$

This function is defined for all real x and sums the probabilities for all outcomes $\leq x$.

Calculating Cumulative Probabilities

- **From a Table:** If you have the probability distribution table for X , then to compute $F(x)$ for each x , you sum up $P(X = k)$ for all $k \leq x$.

Example using the above table:

Properties of the Cumulative Distribution Function

1. **Non-decreasing:** $F(x)$ never decreases as x increases.
2. **Right-continuous:** For discrete variables, $F(x)$ has jump discontinuities at the values of x , but is constant between these values.
3. **Limits:**

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F(x) = 1.$$

6.3 Expected Value of a Discrete Random Variable

The expected value (or mean) of a random variable represents its long-run average if the experiment is repeated many times.

Definition

- **Expected Value $E(X)$:**

$$E(X) = \mu = \sum_x x \cdot P(X = x)$$

This sums each value x weighted by its probability.

Interpretation as the Mean

- $E(X)$ is the center of the probability distribution and is interpreted as the average result if the random experiment were repeated over and over.

Calculation Example

Suppose X has the following distribution:

x	$P(X=x)$
0	0.10
1	0.25
2	0.35
3	0.20
4	0.10

Then,

$$E(X) = 0(0.10) + 1(0.25) + 2(0.35) + 3(0.20) + 4(0.10) = 0 + 0.25 + 0.70 + 0.60 + 0.40 = 1.95.$$

6.4 Variance of a Discrete Random Variable

Variance measures the spread or dispersion of the values around the expected value.

Definition

- **Variance** $Var(X)$:

$$Var(X) = E(X^2) - [E(X)]^2,$$

where

$$E(X^2) = \sum_x x^2 P(X = x).$$

Calculation Example

Using the same distribution as above with $E(X)=1.95$:

1. **Compute** $E(X^2)$:

$$\begin{aligned}
E(X^2) &= 0^2(0.10) + 1^2(0.25) + 2^2(0.35) + 3^2(0.20) + 4^2(0.10). \\
&= 0 + 0.25 + 4(0.35 = 1.4) + 9(0.20 = 1.8) + 16(0.10 = 1.6) = 0 + 0.25 + 1.4 + 1.8 + 1.6 \\
&= 5.05. = 0 + 0.25 + 4(0.35 = 1.4) + 9(0.20 = 1.8) + 16(0.10 = 1.6) \\
&= 0 + 0.25 + 1.4 + 1.8 + 1.6 = 5.05.
\end{aligned}$$

2. Then Compute Variance:

$$Var(X) = 5.05 - (1.95)^2 = 5.05 - 3.8025 = 1.2475 \text{ (approximately).}$$

Alternatively, one can compute $Var(X)$ directly from the definition by finding $(x - \mu)^2$ for each x and its probability, then summing.

6.5 Expected Value and Variance of a Function of XX

Often you encounter a new random variable that is a function of another. For linear functions, the transformation rules simplify the calculation.

Expected Value of a Linear Function

- If $Y = aX + b$ then:

$$E(Y) = E(aX + b) = aE(X) + b.$$

Example: If $E(X) = 2$ and you define $Y = 3X + 4$, then $E(Y) = 3(2) + 4 = 10$.

Variance of a Linear Function

- For the same linear function:

$$Var(Y) = Var(aX + b) = a^2 Var(X).$$

The constant b does not affect the variance. *Example:* If $Var(X) = 1.5$ and $Y = 3X + 4$, then $Var(Y) = 9 \times 1.5 = 13.5$.

These rules are very useful because they allow you to quickly adjust the expected value and variance when applying linear transformations.

6.6 Solving Problems Involving Random Variables

This section is about applying the above principles to real problems.

Using Expected Value to Solve Problems

- **Decision Making:** If you have a game of chance, the expected value tells you the average win (or loss) in the long run.
- **Practical Example:** In a lottery where a ticket can win \\$100 with probability 0.05 and win nothing with probability 0.95, the expected win is:

$$E(\text{winning}) = 100(0.05) + 0(0.95) = 5.$$

Using Variance to Solve Problems

- **Risk and Variability:** Variance provides insight into the risk or uncertainty associated with the outcome.
- **Example:** Comparing two investment returns with the same expected value but different variances; the one with the lower variance is less risky.

Contextual Problems Involving Discrete Random Variables

- You might be given a table, asked to find $E(X)$ and $Var(X)$, or solve a game theory problem.
- **Tip:** Always start by verifying the probability distribution sums to 1, then compute weighted averages for $E(X)$ and follow with variance calculations.

6.7 Using Discrete Uniform Distribution as a Model

The discrete uniform distribution applies when all outcomes in a finite sample space are equally likely.

Conditions for a Discrete Uniform Distribution

- There are a finite number n of outcomes.
- Each outcome has the same probability.

Probability Function

- For each outcome x :

$$P(X = x) = \frac{1}{n}.$$

Expected Value

- The expected value of a discrete uniform distribution is given by:

$$E(X) = \frac{n + 1}{2}.$$

Example: For a fair die ($n=6$):

$$E(X) = \frac{6 + 1}{2} = 3.5.$$

Variance

- The variance is given by:

$$\text{Var}(X) = \frac{(n + 1)(n - 1)}{12}.$$

Example: For a fair die:

$$\text{Var}(X) = \frac{(7)(5)}{12} \approx 2.92.$$

Real-world Application

- **Example:** In board games or lotteries where every outcome (like dice rolls or draws) is equally likely, the discrete uniform distribution is the natural model.
- **Interpretation:** Knowing the mean and variance helps in risk analysis, game fairness assessment, and simulation models.

Chapter Review 6

- **6.1 Discrete Random Variables:** Understand what a random variable is. Its value comes from a sample space, and its behavior is summarized by its probability distribution—which can be represented in tables and diagrams. The discrete uniform distribution is a special case when all outcomes are equally likely.
- **6.2 Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x)$ gathers the cumulative probabilities. Use it to understand the distribution's buildup and remember its properties (non-decreasing, right-continuous, limits 0 and 1).
- **6.3 Expected Value:** The mean $E(X)$ is computed as $\sum xP(X = x)$. It is the long-run average outcome.
- **6.4 Variance:** Variance, calculated via either $\sum (x - \mu)^2 P(X = x)$ or $E(X^2) - [E(X)]^2$, measures data spread.
- **6.5 Linear Functions of X:** Use the rules $E(aX + b) = aE(X) + b$ and $\text{Var}(aX + b) = a^2\text{Var}(X)$ to simplify analysis of transformed variables.
- **6.6 Solving Practical Problems:** Apply these concepts to analyze risks, predict outcomes, and solve real-world problems in games, finance, and more.
- **6.7 Discrete Uniform Distribution as a Model:** When outcomes are equally likely, use the formulas $1/nP(X = x) = 1/n$, $E(X) = (n + 1)/2$, and $\text{Var}(X) = ((n + 1)(n - 1))/12$ to model and analyze experiments like dice rolls.

CHAPTER 7: THE NORMAL DISTRIBUTION

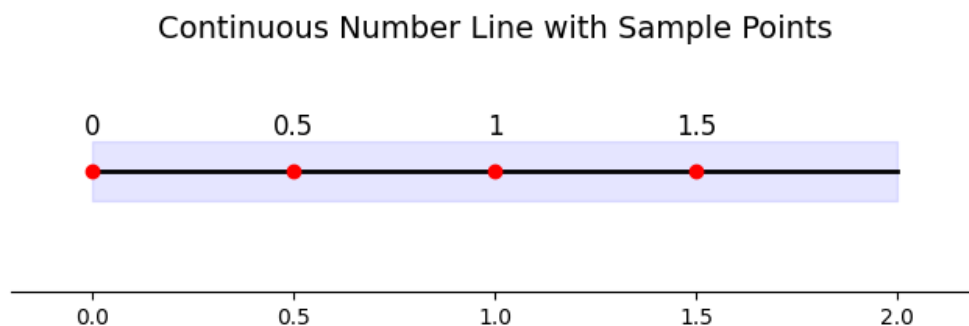
7.1 The Normal Distribution

In this section, we explore the central concepts of the normal distribution by first understanding the underlying ideas of continuous random variables and continuous probability distributions. We then delve into the specifics of the normal distribution, its parameters, and its properties.

A. Continuous Random Variable

Definition and Explanation

A **continuous random variable** is one that can take any value within a given interval—or even an unlimited range—unlike a discrete variable, which is restricted to separate, distinct values. Because the set of possible outcomes is uncountably infinite, the probability of the variable taking on any one exact value is zero. Instead, probabilities are assigned to intervals of values.

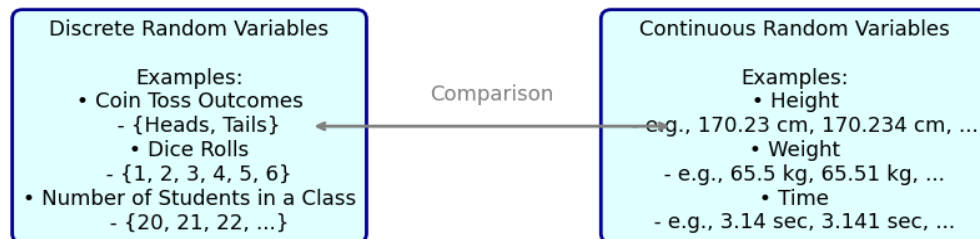


- **Key Points:**
 - **Infinite Possibilities:** The variable can assume any value within the interval (e.g., any real number between 0 and 1, or any height measurement).
 - **Examples:**
 - **Height:** 170.23 cm, 170.234 cm, ...
 - **Weight:** 65.5 kg, 65.51 kg, ...
 - **Temperature:** 22.3°C, 22.3001°C, etc.
 - **Time:** 3.14 seconds, 3.141 seconds, etc.

Example:

Imagine measuring the time it takes for a chemical reaction to complete. Unlike counting the number of leaves on a tree (a discrete countable outcome), time can be any decimal value (e.g., 5.632 seconds). Although you can record time only up to a certain decimal place, theoretically, the value can be infinitely precise.

Diagram 2: Discrete vs Continuous Random Variables



B. Continuous Probability Distribution

Definition and Concept

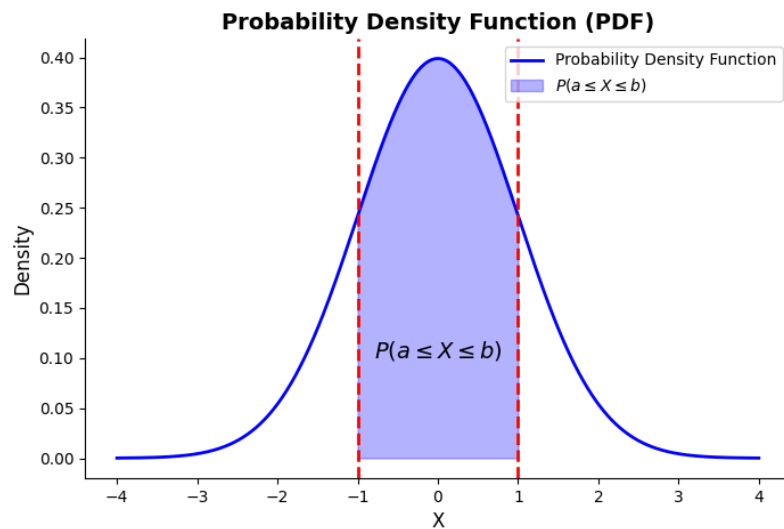
A **continuous probability distribution** describes how the probability is distributed over the values of a continuous random variable. For such variables, the probability of obtaining any specific value is zero. Instead, we define the probability across an interval of values, which is computed as the area under the probability density function (pdf) over that interval.

- **Probability Density Function (pdf):** A function $f(x)$ that describes the likelihood of the random variable taking on a specific value. The probability that the variable X is within an interval $[a, b]$ is given by:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- **Total Area Under the Curve:** The integral of the pdf over all possible values is equal to 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



This represents the certainty that the variable will assume a value within the defined range.

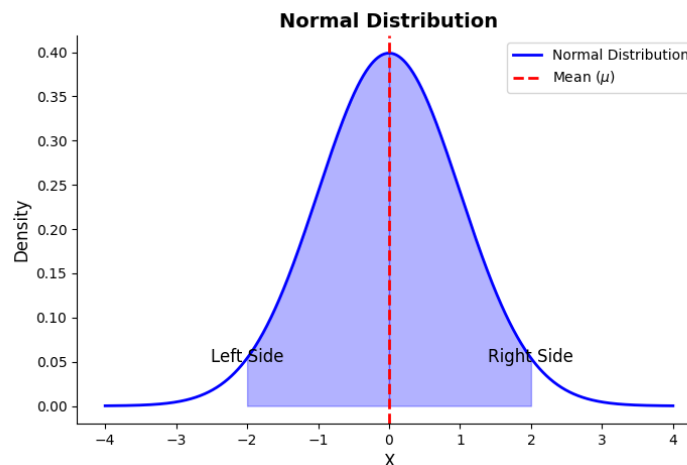
Example:

If X represents the time (in minutes) to complete a task with a continuous probability distribution, the probability that the task takes between 10 and 12 minutes is determined by calculating the area under the pdf from $x = 10$ to $x = 12$.

C. Normal Distribution

The **normal distribution** is arguably the most significant continuous probability distribution in statistics due to its ubiquitous appearance in natural phenomena and measurement errors. It is characterized by its iconic bell-shaped, symmetrical curve.

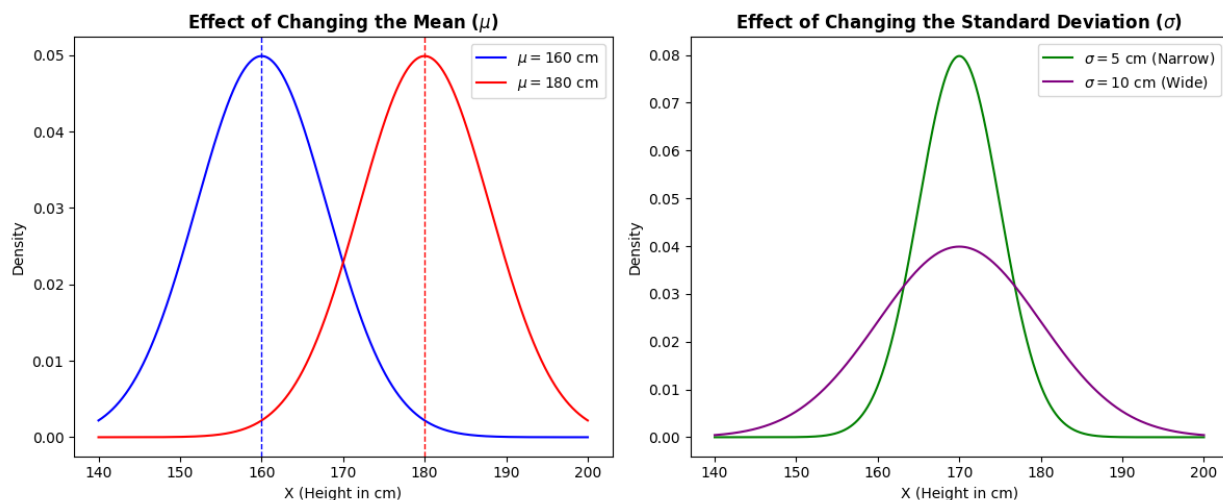
1. Characteristics and Description



- **Bell-Shaped and Symmetrical Curve:** The normal distribution is defined by its smooth, symmetric bell-shaped curve. The symmetry implies that the left half is a mirror image of the right half.
 - **Key Feature:** The distribution is perfectly symmetric about the mean.
- **Mean = Median = Mode:** Because of its symmetry, the central tendency measures—mean, median, and mode—are identical and all located at the center of the distribution.
- **Total Area Under the Curve Equals 1:** The total area under the standard normal curve represents the sum of all probabilities, which is 1.
- **Asymptotic Nature:** The tails of the normal distribution extend infinitely in both directions (positive and negative), approaching the horizontal axis but never touching it. This indicates that every value, however extreme, has a nonzero probability (albeit extremely small).
- **Points of Inflection:** The curve changes its curvature at two points, located one standard deviation from the mean, specifically at $\mu - \sigma$ and $\mu + \sigma$. These are the inflection points where the curvature transitions from concave to convex.
- **Empirical Rule (68–95–99.7 Rule):** This rule provides a quick summary of how data are spread in a normal distribution:
 - Approximately **68%** of the data falls within one standard deviation of the mean.
 - Approximately **95%** of the data falls within two standard deviations.
 - Approximately **99.7%** of the data falls within three standard deviations.

2. Parameters of the Normal Distribution

The exact shape and position of a normal distribution curve are completely determined by two parameters:



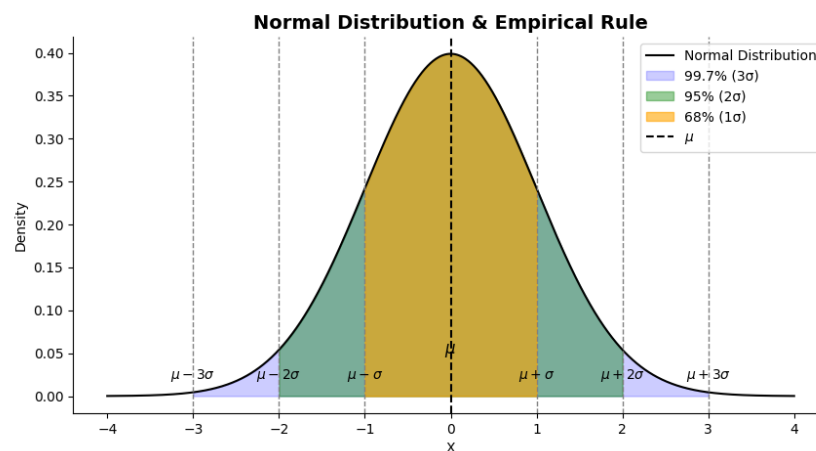
- **Mean (μ):**
 - **Definition:** The mean represents the central or average value of the distribution.
 - **Role in the Distribution:** It determines the center of the bell curve.
 - **Example:** For a dataset representing adult heights, if $\mu = 170$ cm, the peak of the distribution is at 170 cm.

- **Variance (σ^2) and Standard Deviation (σ):**
 - **Variance (σ^2)** measures the spread or dispersion around the mean.
 - **Standard Deviation (σ)** is the square root of the variance and gives a typical measure of deviation from the mean.
 - **Example:** If the variance of exam scores is 25, the standard deviation is $\sigma = 5$. A larger standard deviation implies a wider (flatter) curve, indicating more variability.
- **Notation:** When a random variable X is normally distributed with mean μ and variance σ^2 , we denote it as:

$$X \sim N(\mu, \sigma^2)$$

3. Properties of the Normal Distribution Curve

- **Bell-Shaped and Symmetrical:** The normal curve is smooth, symmetric around the mean μ , and has exactly half its area on each side of the mean.
- **Mean = Median = Mode:** Owing to symmetry, these three measures coincide.
- **Total Area Under the Curve is 1:** This is a requirement of any probability distribution, ensuring that the total probability is 100%.
- **Asymptotic to the x-axis:** The tails extend infinitely toward both positive and negative directions, never touching the x-axis. This implies that no value is impossible, though extreme values have very low probabilities.
- **Points of Inflection:** Occur at $\mu \pm \sigma$; these points mark where the curvature of the graph changes.
- **Empirical Rule (68–95–99.7):**
 - 68% of the values lie within one standard deviation of the mean ($\mu - \sigma$ to $\mu + \sigma$).
 - 95% lie within two standard deviations ($\mu - 2\sigma$ to $\mu + 2\sigma$).
 - 99.7% lie within three standard deviations ($\mu - 3\sigma$ to $\mu + 3\sigma$).



D. Summary

To summarize:

- **Continuous Random Variable:** A variable capable of assuming any value within an interval, providing an unbroken spectrum of possible outcomes. Examples include height, weight, time, and temperature. *Diagram Reminder:* Number line showing the continuum of values.
- **Continuous Probability Distribution:** Defined by a probability density function (pdf), where individual points have zero probability and only intervals (areas under the curve) have meaningful probabilities. The total area under the pdf equals 1. *Diagram Reminder:* Shaded area under a generic pdf.
- **Normal Distribution:** A special continuous distribution that is bell-shaped and symmetric. Its characteristics are entirely determined by its mean (μ) and variance (σ^2).
 - **Key properties include:**
 - **Symmetry:** Mean, median, and mode coincide.
 - **Total probability (area) is 1.**
 - **The tails are asymptotic to the x-axis.**
 - **Inflection points occur at $\mu \pm \sigma$.**
 - **Empirical Rule (68-95-99.7):** Approximately 68% of values fall within one standard deviation, 95% within two, and 99.7% within three standard deviations. *Diagram Reminders:*
 - A clearly labeled bell curve showing the center, inflection points, and shaded empirical rule intervals.
 - Side-by-side diagrams displaying the effect of different parameters.

CHAPTER 7.2: USING TABLES TO FIND PROBABILITIES OF THE STANDARD NORMAL DISTRIBUTION Z

(Reference: [3, p. 155–156])

In this section, we explain the standard normal distribution and illustrate how to use its corresponding tables to find cumulative probabilities. We also detail the process of standardizing any normally distributed variable. The ultimate goal is to empower you to quickly compute probabilities for any normal distribution by referencing a single, standard table.

1. Standard Normal Distribution

Definition and Key Attributes

- **Special Case:** The standard normal distribution is a specific form of the normal distribution defined by a mean $\mu = 0$ and a standard deviation $\sigma = 1$ (thus the variance $\sigma^2 = 1$).
 - **Notation:**

$$Z \sim N(0,1)$$

- **Purpose:** By standardizing any normal variable to have these parameters, we can universally apply one table—the standard normal table—to our probability calculations.

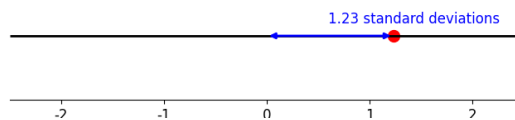
Diagram Suggestion 1

- **Location:** Immediately after this explanation
- **Content:** A bell-shaped curve labeled “Standard Normal Distribution” with the horizontal axis marked $Z = 0$ at the center. Use decorations such as:
 - A vertical line through $Z = 0$ (the mean).
 - Arrows displaying the symmetric nature of the curve on either side.
 - Notations stating “ $\mu = 0$ ” at the center and “ $\sigma = 1$ ” along the horizontal axis.

2. Standard Normal Variable (Z)

Definition and Explanation

- **Standard Normal Variable:** Any value drawn from the standard normal distribution is called a **z-score** or standard normal variable.
 - **Interpretation:** The z-score tells you how many standard deviations a particular value is from the mean. For example, $Z = 1.23$ means that the value is 1.23 standard deviations above the mean of 0.
- **Example:** Consider a data point with a z-score of 1.23. This indicates that this value lies 1.23 standard deviations to the right of the mean.



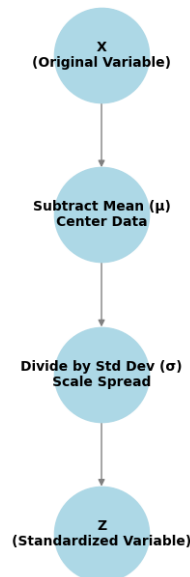
3. Standardizing a Normal Variable (X)

The Process of Standardization

- **Objective:** Convert a normally distributed variable X (with mean μ and standard deviation σ) into a standard normal variable Z . This transformation allows you to use the standard normal table for any normal distribution.
- **Formula:**

$$Z = \frac{X - \mu}{\sigma}$$

- **Step-by-Step Example:** Suppose X represents exam scores where $\mu = 70$ and $\sigma = 10$. If a student scores $X = 85$:
 1. Subtract the mean: $85 - 70 = 15$
 2. Divide by the standard deviation: $\frac{15}{10} = 1.5$ Thus, the standardized score is $Z = 1.5$.



4. Standard Normal Distribution Tables

What They Are and How They Work

- **Definition:** Standard normal distribution tables are pre-calculated tables that list the cumulative probabilities $P(Z < z)$ for various z -values. The entry in the table gives the area under the standard normal curve to the left of that z -value.
- **Structure of the Table:**
 - **Rows:** Represent the integer and the first decimal part of the z -value (e.g., 0.0, 0.1, 0.2, etc.).
 - **Columns:** Represent the second decimal digit (from 0.00 up to 0.09).
- **Usage:** To look up $P(Z < z)$, split z into its two parts (e.g., for $z = 1.23$, use row “1.2” and column “0.03”) and find the intersection in the table.

5. Using the Tables to Find $P(Z < z)$

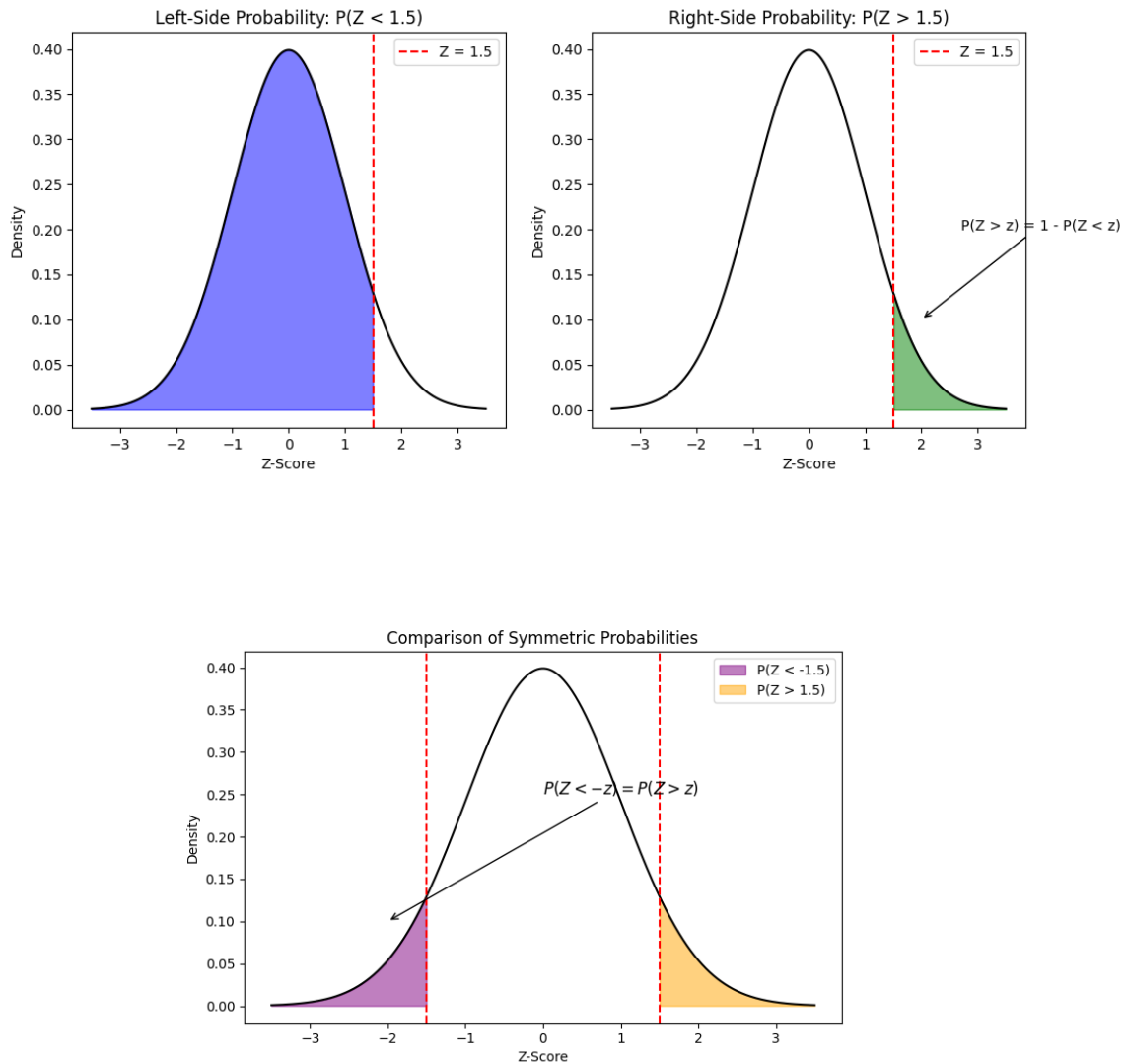
Step-by-Step Procedure

1. **Break Down the z -value:** For a given z (say, $z = 1.23$):
 - Identify the row for the integer part and first decimal: here, 1.2.

- Identify the column for the second decimal: here, 0.03.
2. **Locate the Probability:** The cell where the row and column intersect gives $P(Z < 1.23)$.
- **Example:** Suppose the table indicates 0.8907 at this intersection. Then,

$$P(Z < 1.23) \approx 0.8907$$

6. Using Symmetry Properties to Find Other Probabilities



Understanding and Applying Symmetry

The standard normal distribution's symmetry around $Z=0$ allows for simple relationships to compute probabilities for values not directly listed in the standard table.

- **Right-Tail Probability:**

$$P(Z > z) = 1 - P(Z < z)$$

- **Example:** If $P(Z < 1.23) = 0.8907$, then:

$$P(Z > 1.23) = 1 - 0.8907 = 0.1093$$

- **Left-Tail for Negative z-values:**

$$P(Z < -z) = 1 - P(Z < z)$$

- **Example:** $P(Z < -1.23) = 1 - P(Z < 1.23) = 1 - 0.8907 = 0.1093$ Alternatively, because of symmetry, $P(Z < -1.23)$ equals $P(Z > 1.23)$.
- **Complement for Negative Values:**

$$P(Z > -z) = P(Z < z)$$

This equality holds because the area to the left of a positive z equals the area to the right of its corresponding negative z .

- **Probability Between Two Values:** To find the probability that Z lies between two values a and b , use:

$$P(a < Z < b) = P(Z < b) - P(Z < a)$$

- **Example:** For $0.50 < Z < 1.23$, if the table gives $P(Z < 1.23)$ and $P(Z < 0.50)$, then subtract the latter from the former to obtain the probability between these limits.

Summary

In summary:

- **Standard Normal Distribution:** A fundamental distribution defined by $\mu = 0$ and $\sigma = 1$; it simplifies calculations by providing a universal scale.
- **Standard Normal Variables (Z-scores):** Every value of Z represents the number of standard deviations it is away from the mean. This allows for direct comparisons across different datasets.
- **Standardization Process:** Any normal variable X with mean μ and standard deviation σ can be converted to the standard normal form using:

$$Z = \frac{X - \mu}{\sigma}$$

This transformation is key to using the standard normal table for probability calculations.

- **Using the Standard Normal Table:** The tables provide cumulative probabilities $P(Z < z)$. By splitting a z-value into its two components (rows and columns), we can quickly find the corresponding area under the curve.
- **Symmetry Properties:** Due to the symmetric nature of the standard normal distribution:
 - $P(Z > z) = 1 - P(Z < z)$
 - $P(Z < -z) = P(Z > z)$
 - The probability between two values is determined by the difference $P(Z < b) - P(Z < a)$.

CHAPTER 7.3: USING TABLES TO FIND THE VALUE OF z GIVEN A PROBABILITY

(Reference: [3, p. 158])

In many statistical problems, you are not given a z-value but rather a probability. The goal in such cases is to find the z-value that corresponds to that cumulative probability. This process is essentially the reverse of using the z-table to find a probability from a given z-value and is foundational in hypothesis testing and confidence interval estimation.

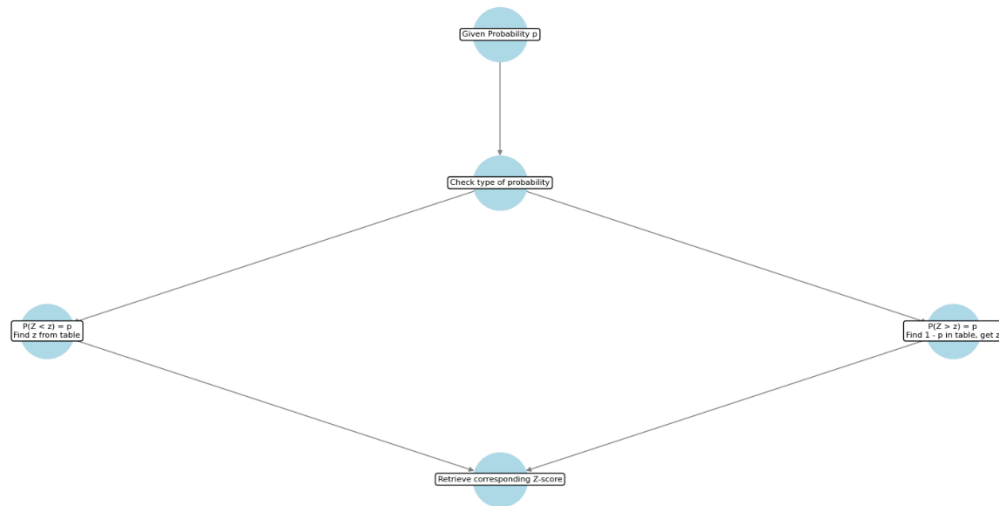
1. Overview and Fundamental Concepts

1.1 The Reverse Lookup Problem

- **Concept:** Instead of determining the probability $P(Z < z)$ for a given z-score, you are asked to find the z-score when you are given a probability.
- **Example Situation:** Imagine you're told that 89.07% of observations fall below a certain value in a standard normal distribution. You must determine the z-value, call it z^* , such that:

$$P(Z < z^*) = 0.8907$$

- **Purpose:** This reverse lookup is essential when:
 - Setting critical values for hypothesis tests.
 - Constructing confidence intervals.
 - Converting confidence levels to z-scores.



2. Finding z Given $P(Z < z) = p$

2.1 Step-by-Step Procedure

When presented with a left-tail probability p , the process is as follows:

1. **Locate the Probability in the Table:** Search within the standard normal distribution table for the probability value that is equal to (or very near) p .
 - **Tip:** The table is divided into rows and columns. Rows represent the z -value rounded to the first decimal (e.g., 1.2) and columns represent the second decimal portion (e.g., 0.03).
2. **Read off the z -value:** Once you find the probability closest to p , note the corresponding z -value by combining the row header (integer and first decimal) and the column header (second decimal).
 - **Example:** Suppose you are given $P(Z < z) = 0.8907$. You then locate 0.8907 in the table, which might appear at the intersection of the row labeled “1.2” and the column labeled “0.03”. Thus, the corresponding z -value is:

$$z = 1.23$$

2.2 In-Depth Example

- **Scenario:** Assume you are provided with $P(Z < z) = 0.9750$.
- **Process:**
 1. Scan the body of the z -table to locate the value 0.9750 or the closest approximation.
 2. You find that 0.9750 corresponds to the intersection of the row “1.9” and the column “0.05.”
- **Result:** Thus, the z -value is:

$$z = 1.95$$

- **Interpretation:** This tells you that 97.50% of the data in a standard normal distribution lie below a z-score of 1.95.

3. Finding z for Upper-Tail Probabilities $P(Z > z) = p$

3.1 Understanding the Complement Rule

- **Concept:** The standard normal table is typically arranged to show cumulative probabilities from the left tail. When you are given a probability in the upper tail, you must convert it to the corresponding left-tail probability.
- **Complement Rule:**

$$P(Z < z) = 1 - P(Z > z)$$

- **Procedure:**
 1. **Given:** $P(Z > z) = p$
 2. **Compute:** $P(Z < z) = 1 - p$
 3. Use the standard normal table to find the z-value that corresponds to $1 - p$.

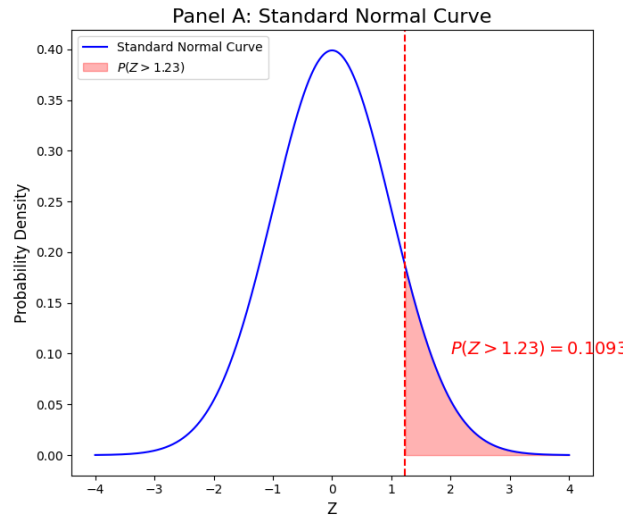
3.2 Example

- **Scenario:** Suppose you are told that $P(Z > z) = 0.1093$.
- **Steps:**
 1. **Convert to Left-Tail Probability:**

$$P(Z < z) = 1 - 0.1093 = 0.8907$$

2. **Lookup:** Use the table as outlined in Section 2 to find the z-value for 0.8907, which as previously seen is approximately 1.23.
- **Result:** Thus, the corresponding z-value is:

$$z = 1.23$$



Panel B: Z-table Lookup

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.1549	0.1665	0.1785	0.1905	0.2025	0.2146	0.2266	0.2387	0.2508	0.2629
0.1	0.2743	0.2859	0.2975	0.3090	0.3206	0.3320	0.3438	0.3544	0.3659	0.3770
0.2	0.3888	0.3995	0.4102	0.4209	0.4315	0.4422	0.4528	0.4633	0.4738	0.4841
0.3	0.4944	0.5044	0.5143	0.5241	0.5339	0.5436	0.5532	0.5628	0.5724	0.5819
0.4	0.5914	0.6011	0.6107	0.6203	0.6299	0.6394	0.6488	0.6582	0.6675	0.6769
0.5	0.6861	0.6954	0.7046	0.7139	0.7231	0.7324	0.7416	0.7508	0.7599	0.7691
0.6	0.7782	0.7874	0.7965	0.8056	0.8146	0.8236	0.8325	0.8414	0.8502	0.8590
0.7	0.8678	0.8765	0.8852	0.8938	0.9024	0.9109	0.9193	0.9277	0.9360	0.9442
0.8	0.9525	0.9607	0.9688	0.9768	0.9847	0.9925	1.0002	1.0079	1.0155	1.0231
0.9	1.0307	1.0382	1.0456	1.0530	1.0604	1.0677	1.0749	1.0821	1.0893	1.0964
1.0	1.1035	1.1105	1.1174	1.1243	1.1311	1.1379	1.1446	1.1513	1.1580	1.1646
1.1	1.1712	1.1778	1.1843	1.1908	1.1973	1.2037	1.2101	1.2165	1.2228	1.2291
1.2	1.2354	1.2417	1.2479	1.2541	1.2603	1.2665	1.2726	1.2787	1.2847	1.2907
1.3	1.2967	1.3026	1.3085	1.3144	1.3203	1.3261	1.3319	1.3377	1.3435	1.3492
1.4	1.3549	1.3606	1.3663	1.3719	1.3775	1.3831	1.3887	1.3942	1.3997	1.4052
1.5	1.4106	1.4161	1.4215	1.4269	1.4323	1.4377	1.4430	1.4483	1.4536	1.4589
1.6	1.4641	1.4693	1.4745	1.4796	1.4847	1.4898	1.4948	1.4998	1.5048	1.5098
1.7	1.5147	1.5196	1.5245	1.5294	1.5342	1.5390	1.5438	1.5486	1.5534	1.5581
1.8	1.5628	1.5675	1.5722	1.5768	1.5814	1.5859	1.5905	1.5950	1.5995	1.6040
1.9	1.6084	1.6129	1.6173	1.6217	1.6261	1.6305	1.6349	1.6392	1.6435	1.6478
2.0	1.6520	1.6562	1.6604	1.6646	1.6687	1.6728	1.6769	1.6809	1.6849	1.6888
2.1	1.6927	1.6967	1.7006	1.7045	1.7083	1.7122	1.7160	1.7198	1.7235	1.7273
2.2	1.7311	1.7347	1.7384	1.7420	1.7455	1.7490	1.7525	1.7560	1.7594	1.7628
2.3	1.7663	1.7696	1.7729	1.7764	1.7798	1.7831	1.7864	1.7897	1.7930	1.7962
2.4	1.7995	1.8027	1.8059	1.8091	1.8123	1.8154	1.8186	1.8217	1.8248	1.8279
2.5	1.8309	1.8340	1.8370	1.8400	1.8430	1.8460	1.8489	1.8518	1.8547	1.8576
2.6	1.8605	1.8634	1.8663	1.8691	1.8720	1.8749	1.8777	1.8805	1.8834	1.8862
2.7	1.8890	1.8918	1.8946	1.8974	1.9002	1.9029	1.9057	1.9084	1.9112	1.9139
2.8	1.9166	1.9192	1.9219	1.9245	1.9271	1.9297	1.9322	1.9347	1.9372	1.9397
2.9	1.9422	1.9446	1.9471	1.9495	1.9519	1.9543	1.9567	1.9590	1.9614	1.9637
3.0	1.9651	1.9674	1.9697	1.9720	1.9742	1.9764	1.9786	1.9808	1.9829	1.9850
3.1	1.9871	1.9892	1.9913	1.9934	1.9954	1.9974	1.9994	2.0014	2.0034	2.0054
3.2	2.0073	2.0093	2.0112	2.0131	2.0150	2.0169	2.0188	2.0206	2.0225	2.0243
3.3	2.0261	2.0279	2.0297	2.0315	2.0332	2.0350	2.0367	2.0384	2.0401	2.0418
3.4	2.0434	2.0451	2.0468	2.0484	2.0501	2.0517	2.0533	2.0549	2.0564	2.0580
3.5	2.0596	2.0611	2.0626	2.0641	2.0656	2.0671	2.0686	2.0700	2.0715	2.0729
3.6	2.0744	2.0758	2.0772	2.0786	2.0801	2.0815	2.0829	2.0843	2.0857	2.0871
3.7	2.0885	2.0898	2.0912	2.0926	2.0939	2.0952	2.0965	2.0978	2.0991	2.1004
3.8	2.1017	2.1030	2.1043	2.1056	2.1068	2.1081	2.1093	2.1105	2.1117	2.1129

4. Using Symmetry for Negative z-Values

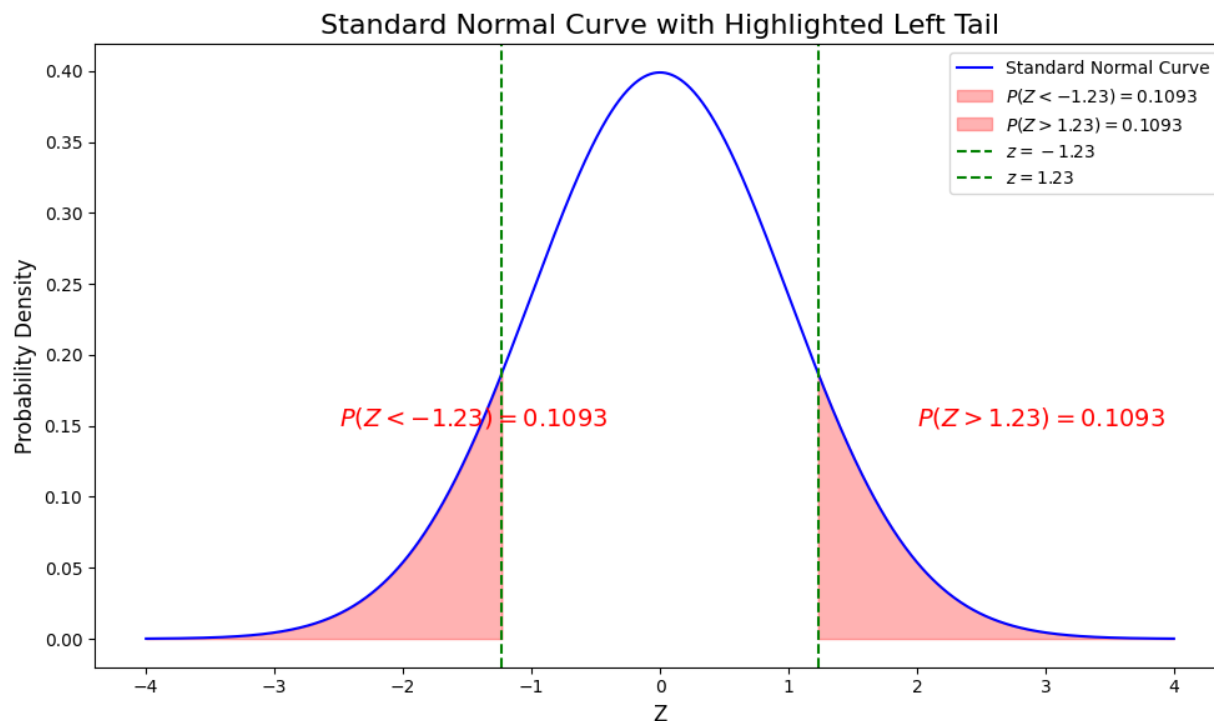
4.1 The Role of Symmetry in the Standard Normal Distribution

- **Symmetry Principle:** The standard normal distribution is perfectly symmetric about zero, meaning the area to the left of a negative z-value is the same as the area to the right of the corresponding positive z-value.
- **Application:** If a given probability pp is less than 0.5, this indicates that the corresponding z-value will be negative.

4.2 Example and Process

- **Scenario:** Find the z-value when $P(Z < z) = 0.1093$.
- **Procedure Using Symmetry:**
 1. **Recognize:** Since 0.1093 is less than 0.5, the z-value will be negative.
 2. **Find the Positive Equivalent:** Look up $P(Z < z')$ where $z' = 1 - 0.1093$ or equivalently use the symmetry property: $P(Z < -z) = P(Z > z)$. From earlier, we saw that $P(Z > 1.23) \approx 1 - 0.8907 = 0.1093$.
 3. **Assign the Negative Sign:** Therefore, the z-value corresponding to $P(Z < z) = 0.1093$ is:

$$z = -1.23$$



5. Summary

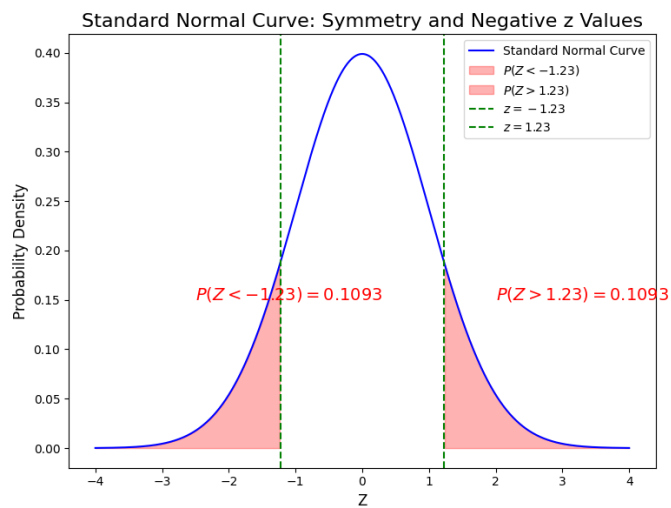
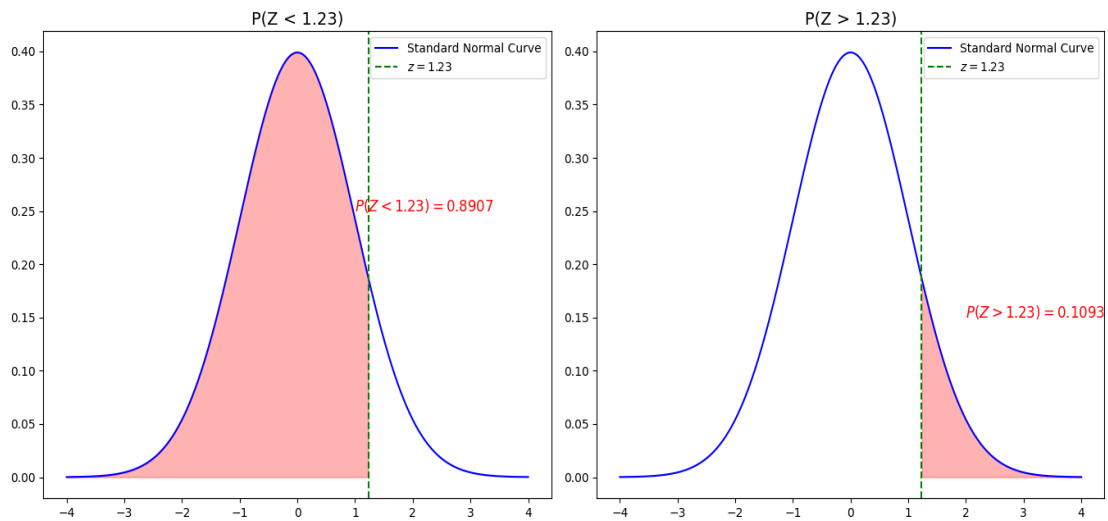
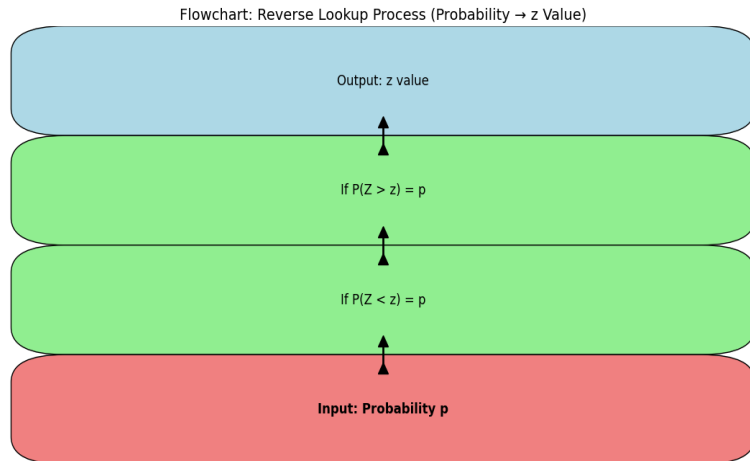
Key Points Recap

- **Reverse Lookup:** When provided with a probability, the goal is to work backward through the standard normal table to find the z-value that defines that cumulative area.
- **Steps for Left-Tail Probability $P(Z < z) = p$:**
 1. Search the standard normal table for the value closest to p .
 2. Read off the z-value at the corresponding row and column.
- **Upper-Tail Probabilities $P(Z > z) = p$:** Convert the right-tail probability to a left-tail probability using:

$$P(Z < z) = 1 - p$$

Then proceed with the standard lookup.

- **Symmetry for Negative Values:** If $p < 0.5$, the z-value is negative; you can find the corresponding positive z-value using the table, then assign a negative sign.



CHAPTER 7.4: THE STANDARD NORMAL DISTRIBUTION

(Reference: [3, p. 161])

This section reinforces the fundamental role of the standard normal distribution as a universal benchmark for analyzing any normally distributed variable. By converting any normal variable into its z-score, we can use one single set of cumulative probability tables to compute probabilities, regardless of the variable's original mean or variance.

1. Concept and Significance of the Standard Normal Distribution

Definition and Overview

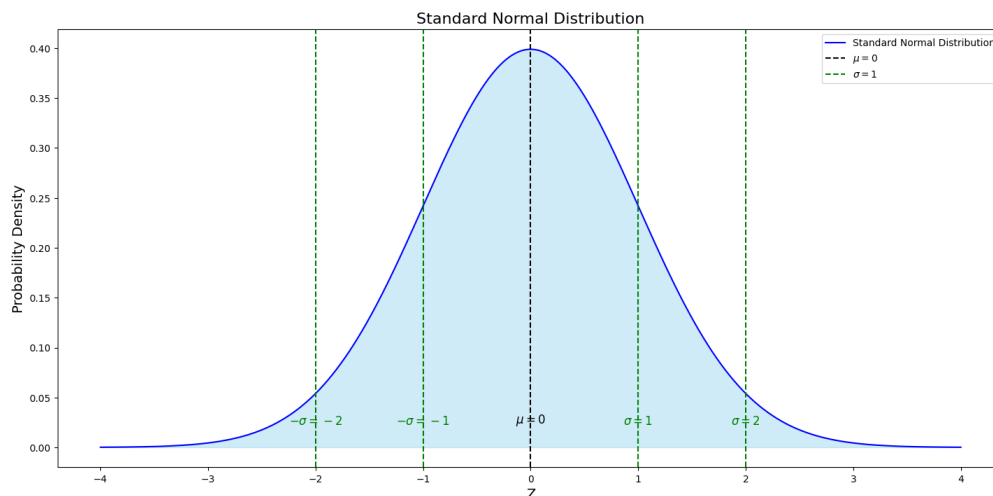
- **Standard Normal Distribution:** This is a special case of the normal distribution defined by a mean, $\mu = 0$, and a standard deviation, $\sigma = 1$ (thus the variance $\sigma^2 = 1$). It is denoted as:

$$Z \sim N(0, 1)$$

- **Purpose:** The primary reason for standardizing a normal variable is to convert various normal distributions (with different means and standard deviations) into one common framework. This process simplifies probability calculations since every normal variable can be transformed into the standard normal form.
- **Key Idea:** Any normally distributed variable $X \sim N(\mu, \sigma^2)$ can be transformed to a standard normal variable Z using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

This transformation linearizes the variable onto a universal scale, allowing the use of standard z-tables.



2. The Process of Standardization

Step-by-Step Explanation

- **The Transformation Formula:** To standardize any normal variable X , use:

$$Z = \frac{X - \mu}{\sigma}$$

Here, subtracting the mean centers the distribution about 0, and dividing by the standard deviation scales the data so that its spread is measured in units of standard deviations.

- **Detailed Example:**

Example 1: Suppose exam scores are normally distributed such that $X \sim N(70, 10^2)$. If a student scores 85, what is the corresponding z-score?

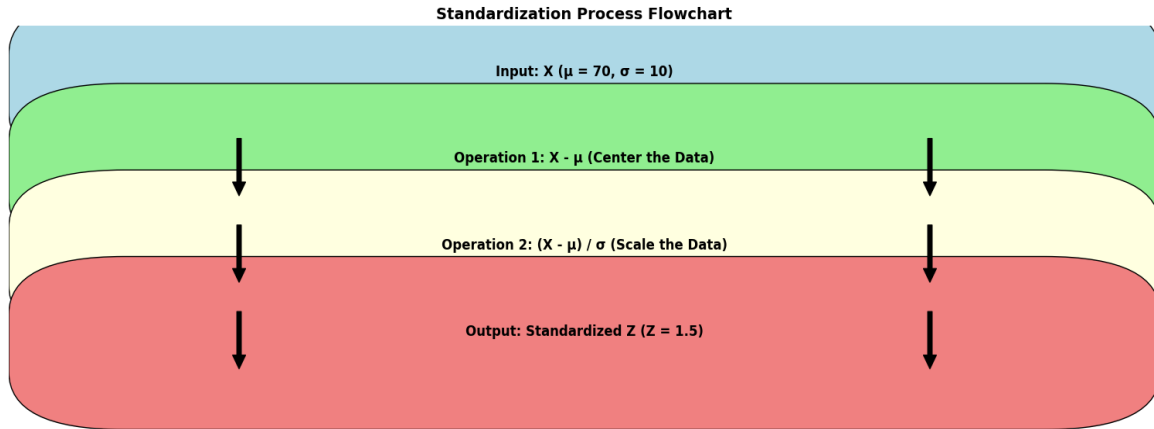
- **Step 1:** Subtract the mean:

$$85 - 70 = 15$$

- **Step 2:** Divide by the standard deviation:

$$Z = \frac{15}{10} = 1.5$$

Interpretation: A score of 85 is 1.5 standard deviations above the mean.

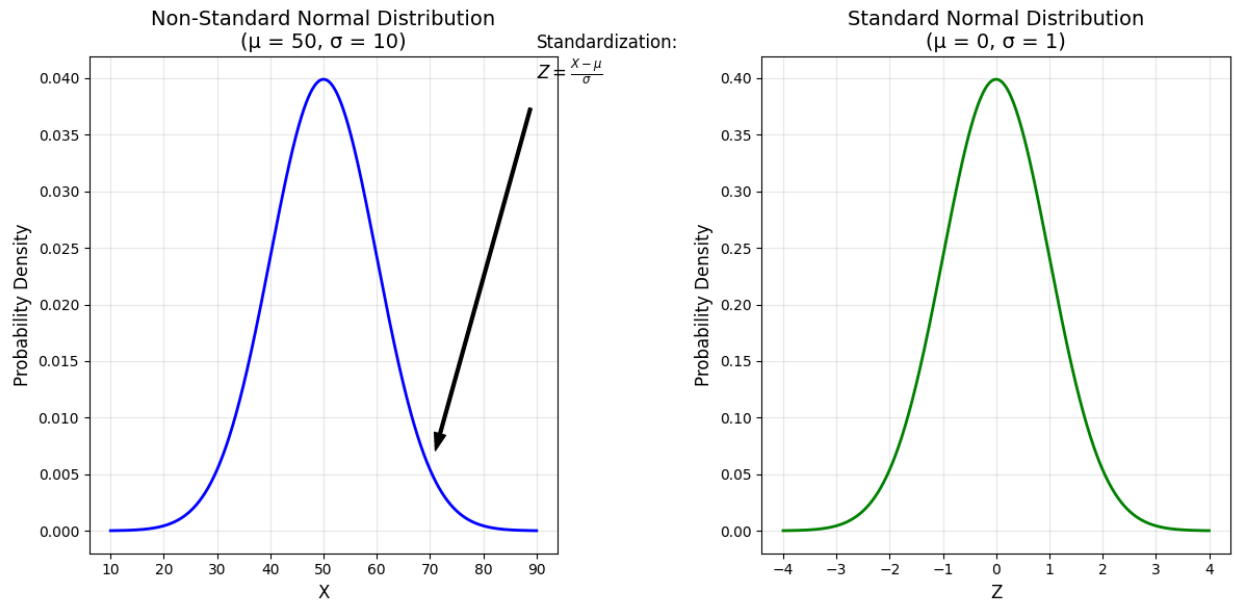


3. Using the Standard Normal Distribution Table

Standardization in Practice

- **How It Works:** Once a variable is standardized, you use a single table to find cumulative probabilities $P(Z < z)$ for any z-score.
- **Example:** With $Z=1.5$ obtained from the previous standardization example, one can look up $P(Z < 1.5)$ in the z-table, which might yield approximately 0.9332. This tells you that 93.32% of the distribution lies below that score.
- **Interpretation and Utility:** The ability to convert to the standard normal form makes it possible to determine probabilities, set critical values, and perform statistical tests using a common reference.

	0.00	0.01	0.02	0.03	0.04	0.05
1.0	0.8413	0.8438	0.8461	0.8484	0.8508	0.8531
1.1	0.8643	0.8667	0.8693	0.8717	0.874	0.8763
1.2	0.8849	0.8879	0.8902	0.8925	0.8948	0.8971
1.3	0.9332	0.9336	0.9352	0.9365	0.9378	0.9391



5. Summary

Key Points Recap

- **Definition:** The standard normal distribution, $Z \sim N(0,1)$, is a normal distribution with a mean of 0 and a standard deviation of 1, serving as a foundational reference for all normal distributions.
- **Standardization Process:** Any normal variable $X \sim N(\mu, \sigma^2)$ can be transformed into a standard normal variable using:

$$Z = \frac{X - \mu}{\sigma}$$

This process centers and scales the data, making it compatible with standard z-tables.

Diagram Recaps

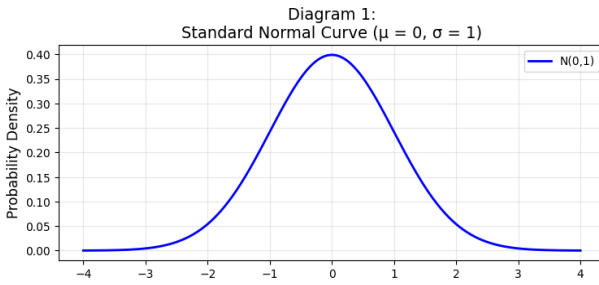
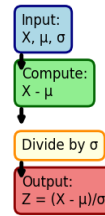


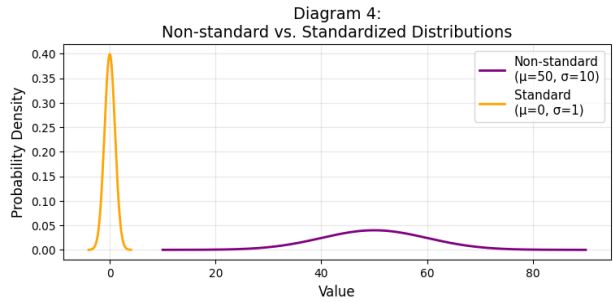
Diagram 2:
Standardization Flowchart



Example:
 $X = 85, \mu = 70, \sigma = 10$
 $\Rightarrow X - \mu = 15$
 $\Rightarrow Z = 15/10 = 1.5$

Diagram 3:
Snapshot of Standard Normal Table

Z	0.00	0.01	0.02	0.03
0.0	0.5000	0.5040	0.5080	0.5120
0.1	0.5398	0.5438	0.5478	0.5517
0.2	0.5793	0.5832	0.5871	0.5910
0.3	0.6179	0.6217	0.6255	0.6293



CHAPTER 7.5: FINDING μ AND σ

(Reference: [3, p. 164])

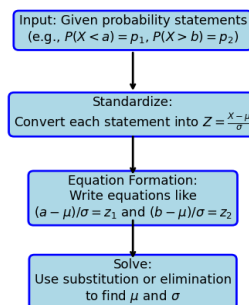
In many real-world applications, the parameters of a normal distribution—the mean μ and the standard deviation σ —are not given directly. Instead, you are often provided with probability information (e.g., cumulative probability statements) that can be used to determine these parameters. In this chapter, we detail a systematic approach to translate probability statements into equations, and then solve for μ and σ .

1. Overview and Concept

1.1 Context and Importance

- Objective:** When μ and/or σ are unknown, available probability statements about the variable X (which is assumed to be normally distributed) are used to create one or more equations. These equations are then solved for the missing parameters.

Diagram 1: Overview Flowchart



2. Translating Probability Information into Equations

2.1 Basic Principle

- **General Approach:**
 - When given a probability statement involving X such as:

$$P(X < x_0) = p$$

you first standardize X using:

$$Z = \frac{X - \mu}{\sigma}.$$

- The probability statement becomes:

$$P\left(Z < \frac{x_0 - \mu}{\sigma}\right) = p.$$

- **Finding the Equivalent z-Value:** Use the standard normal distribution table to find the z-value (z_p) that corresponds to the cumulative probability p. Then you can set up the equation:

$$\frac{x_0 - \mu}{\sigma} = z_p.$$

2.2 Detailed Example

- **Example 1:** Suppose you know that

$$P(X < 50) = 0.8413,$$

and X is normally distributed with unknown μ and σ .

1. **Standardize the Statement:**

$$P\left(Z < \frac{50 - \mu}{\sigma}\right) = 0.8413.$$

2. **Find z-value:** The standard normal table tells us that

$$P(Z < 1) \approx 0.8413$$

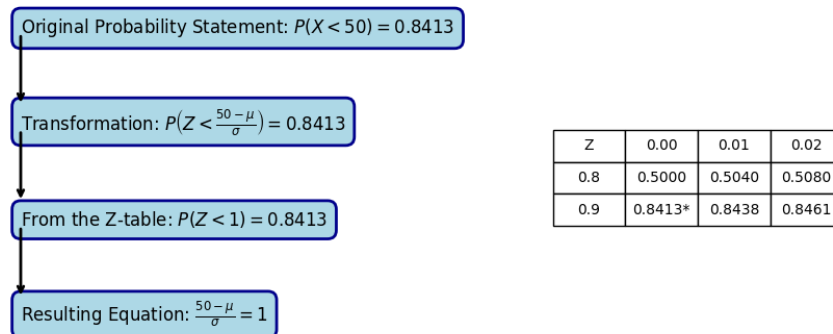
3. **Set Up the Equation:** Therefore, we obtain:

$$\frac{50 - \mu}{\sigma} = 1,$$

which can be rearranged to:

$$50 - \mu = \sigma.$$

Diagram 2: Equation Transformation for Standardization



3. Solving for μ and σ

3.1 Single Probability Statement (One Parameter Unknown)

- **Scenario:** If either μ or σ is known, one probability statement provides enough information to solve for the unknown parameter.
 - **Example:** If σ is known, then from:

$$\frac{50 - \mu}{\sigma} = 1,$$

we solve for μ as:

$$\mu = 50 - \sigma.$$

3.2 Two Probability Statements (Both Parameters Unknown)

- **General Process:** When both μ and σ are unknown, at least two independent probability statements are needed to solve the system.
 - **Example:** Suppose we are given:
 1. $P(X < 10) = 0.2$
 2. $P(X > 15) = 0.1$

Step-by-Step Example

1. **First Probability Statement:** $P(X < 10) = 0.2$

- **Standardize:**

$$P\left(Z < \frac{10 - \mu}{\sigma}\right) = 0.2.$$

- **Using the z-table:** For $p = 0.2$, the corresponding z-value is approximately $z \approx -0.84$ (since $P(Z < -0.84) \approx 0.2$).
- **Equation 1:**

$$\frac{10 - \mu}{\sigma} = -0.84$$

2. **Second Probability Statement:** $P(X > 15) = 0.1$

- **Convert:** Since $P(X > 15) = 0.1$, then

$$P(X < 15) = 1 - 0.1 = 0.9.$$

- **Standardize:**

$$P\left(Z < \frac{15 - \mu}{\sigma}\right) = 0.9.$$

- **From the z-table:** The z-value for 0.9 is approximately $z \approx 1.28$.
- **Equation 2:**

$$\frac{15 - \mu}{\sigma} = 1.28.$$

3. **System of Equations:** We now have:

- Equation (1): $\frac{10 - \mu}{\sigma} = -0.84 \Rightarrow 10 - \mu = -0.84 \sigma$.
- Equation (2): $\frac{15 - \mu}{\sigma} = 1.28 \Rightarrow 15 - \mu = 1.28 \sigma$.

4. **Solving the System:** Subtract Equation (1) from Equation (2):

$$\begin{aligned} (15 - \mu) - (10 - \mu) &= 1.28\sigma - (-0.84\sigma) \\ 15 - 10 &= 1.28\sigma + 0.84\sigma \\ 5 &= 2.12\sigma \Rightarrow \sigma = \frac{5}{2.12} \approx 2.36. \end{aligned}$$

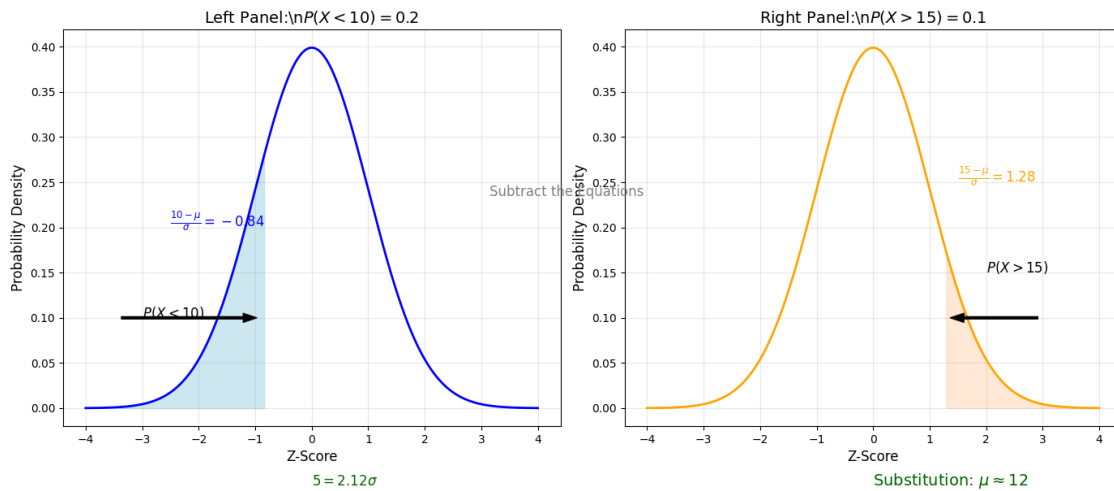
Substitute $\sigma \approx 2.36$ back into Equation (1) (or Equation (2)) to solve for μ :

$$10 - \mu = -0.84 \times 2.36 \approx -1.98 \Rightarrow \mu = 10 + 1.98 \approx 11.98.$$

5. **Result:** The approximate values are:

$$\mu \approx 12 \quad \text{and} \quad \sigma \approx 2.36.$$

Diagram 3: System of Equations for Solving μ and σ



4. Summary

Key Points Recap

- **Purpose:** To determine the unknown mean μ and standard deviation σ of a normal distribution using given probability statements.
- **General Approach:**
 1. **Translate Probability Statements:** Convert statements like $P(X < x_0) = p$ into standardized form:

$$P\left(Z < \frac{x_0 - \mu}{\sigma}\right) = p.$$

2. **Find Corresponding z-Values:** Use the standard normal distribution table to find the z-value z_p associated with p .
3. **Set Up Equations:** Write equations such as:

$$\frac{x_0 - \mu}{\sigma} = z_p.$$

4. **Solve the Equations:** If one parameter is unknown, directly solve the equation. If both μ and σ are unknown, use at least two independent probability statements to form a system of equations and solve by substitution or elimination.
- **Detailed Example Recap:** Using two probability statements— $P(X < 10) = 0.2$ and $P(X > 15) = 0.1$ —we derived a system of equations:

$$\frac{10 - \mu}{\sigma} = -0.84 \quad \text{and} \quad \frac{15 - \mu}{\sigma} = 1.28.$$

Solving these yielded approximate values $\mu \approx 12$ and $\sigma \approx 2.36$.

Diagram 1: Overall Flowchart

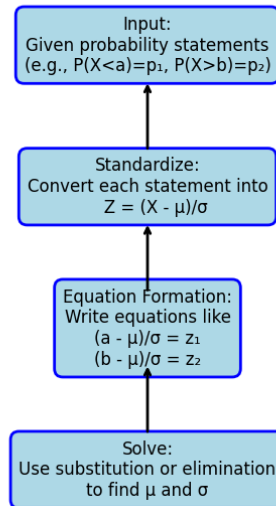


Diagram 2: Transformation Diagram

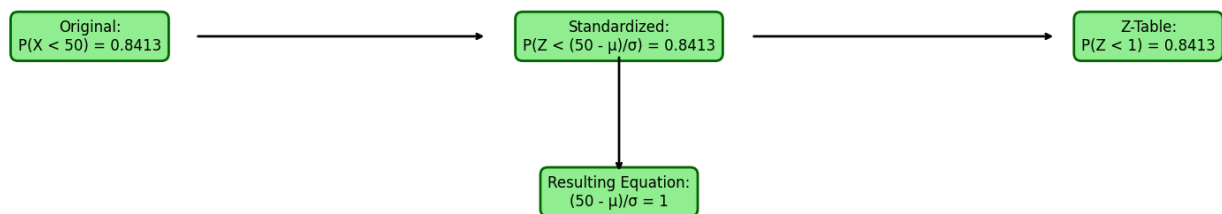


Diagram 3: System of Equations Diagram

